Lab 04

Energy Operations Weather Agent

Microsoft Copilot Studio — Intermediate Lab Walkthrough

Duration: 2.5 hours | Difficulty: Intermediate

Skills Covered: Agent Creation, Topics & Variables, Connectors, Power Fx, Custom Prompts, Connected Agents, Evaluations

Generated: June 2026

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Introduction

In this lab, you will build an Energy Operations Weather Agent that demonstrates how Microsoft Copilot Studio can integrate real-time external data sources with AI-powered analysis to deliver actionable business intelligence.

What you will build:

- A conversational agent that retrieves live weather data via MSN Weather connectors
- A custom AI prompt tool that generates energy grid operations briefings
- A connected child agent (Crew Safety Advisor) for specialized crew safety recommendations
- An evaluation test suite to validate agent quality at scale

Why this matters:

Energy utilities must make rapid operational decisions based on weather conditions. This agent demonstrates how AI can transform raw weather data into structured operations intelligence — reducing decision latency from hours to seconds.

Prerequisites

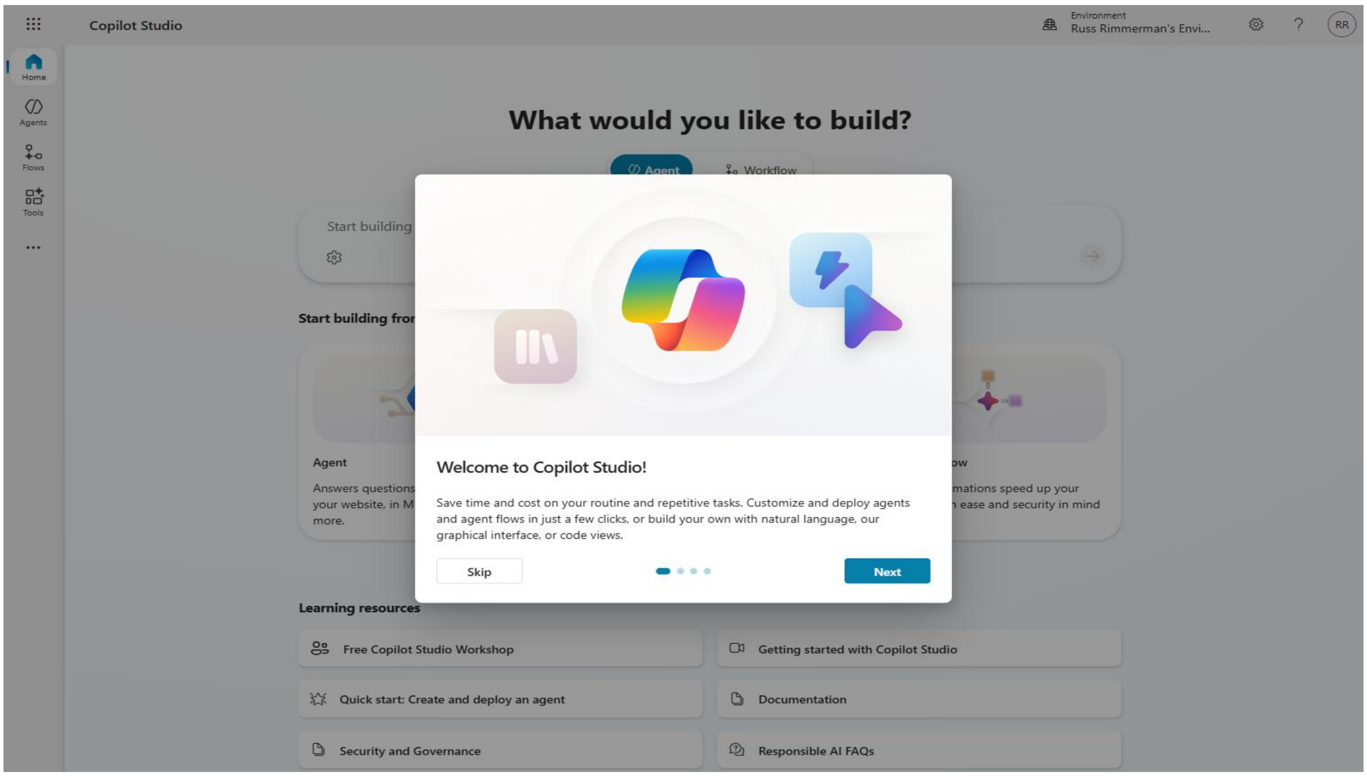
- Microsoft Copilot Studio access (copilotstudio.microsoft.com)
- Power Platform environment with Dataverse
- MSN Weather connector access (included with most M365 licenses)
- Basic familiarity with Copilot Studio interface (Lab 01-03 recommended)

Section 1: Create the Agent

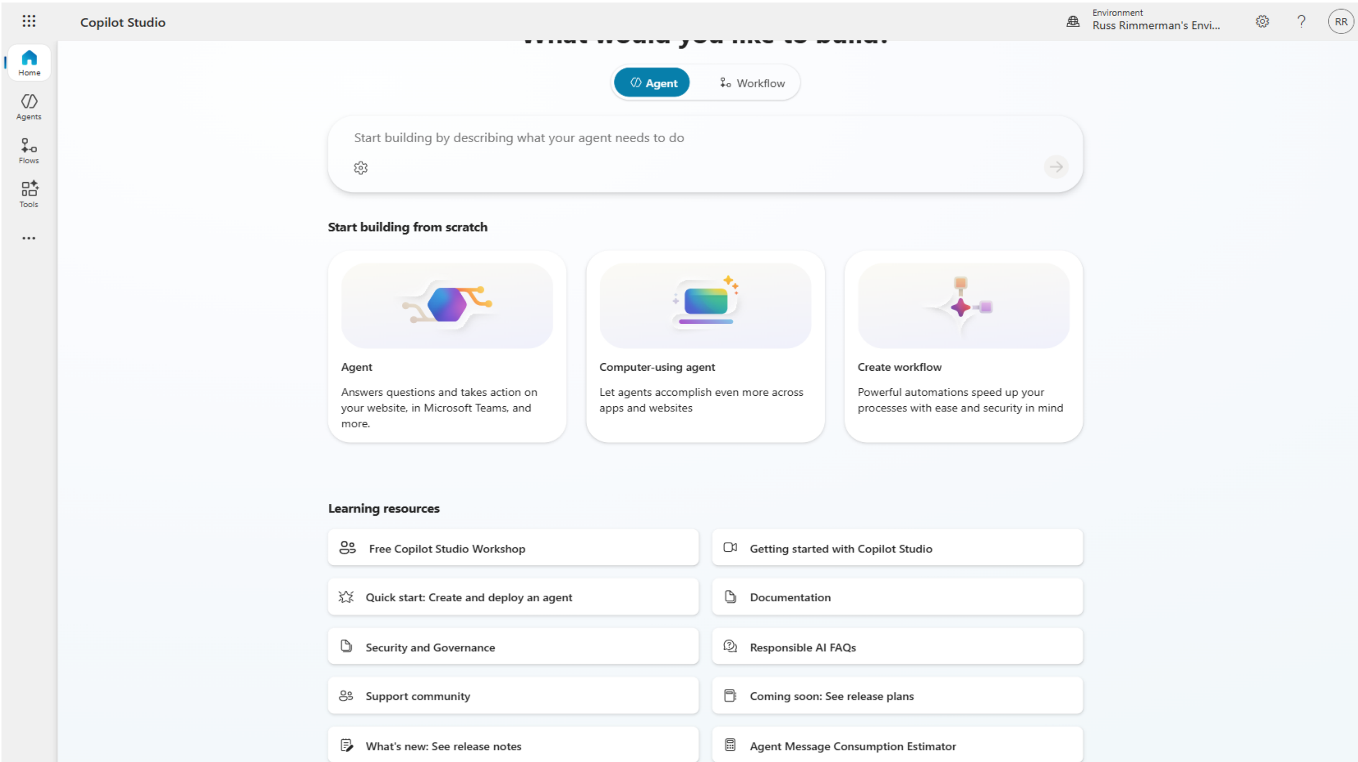
Every Copilot Studio solution starts with creating an agent. The agent is the top-level container that holds your topics, tools, knowledge sources, and configuration. In this section, you will create the Energy Operations Weather Agent with specialized instructions that guide the AI's behavior for energy grid scenarios.

1.1 Navigate to Copilot Studio

Step 1: Open your browser and navigate to copilotstudio.microsoft.com. Ensure you are signed into your Power Platform environment.

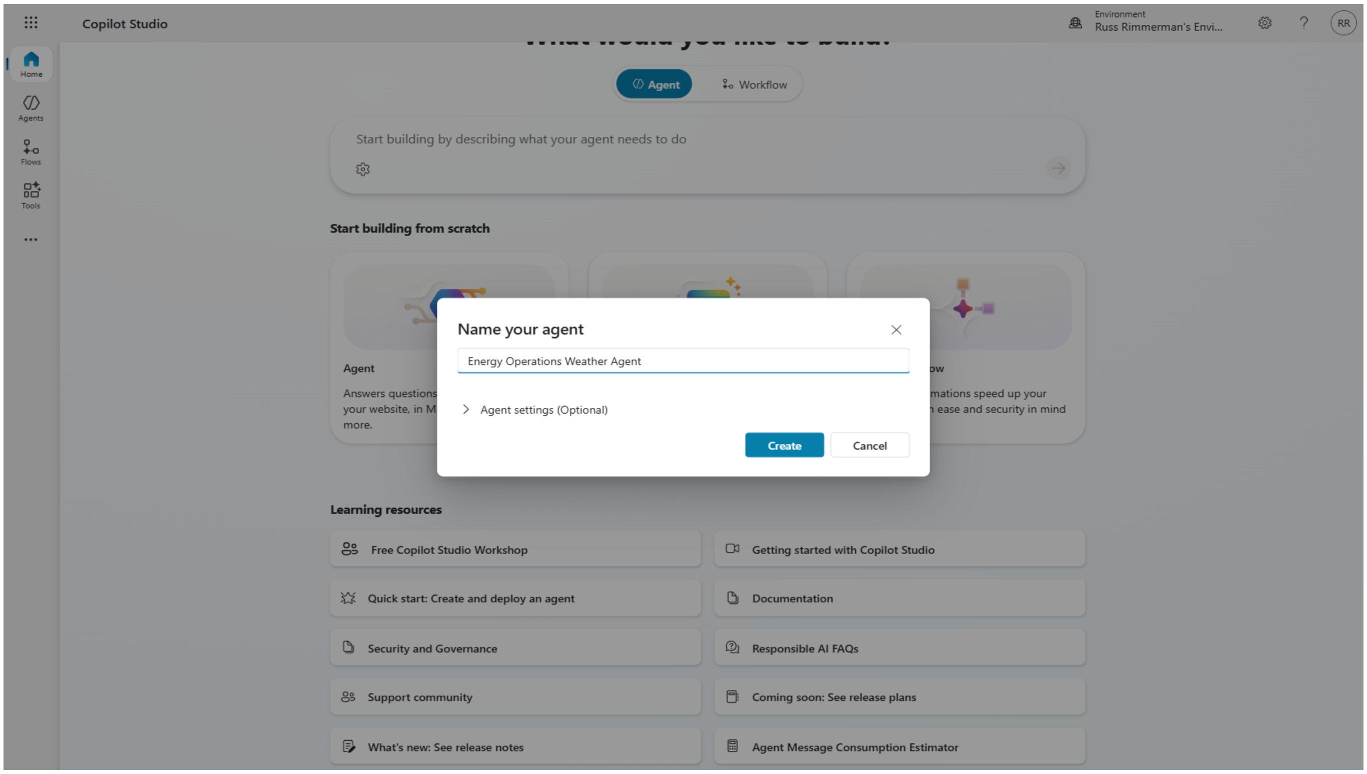


Step 2: From the Home page, click "New agent" or select "Agents" from the left navigation. You'll see the agent creation interface.



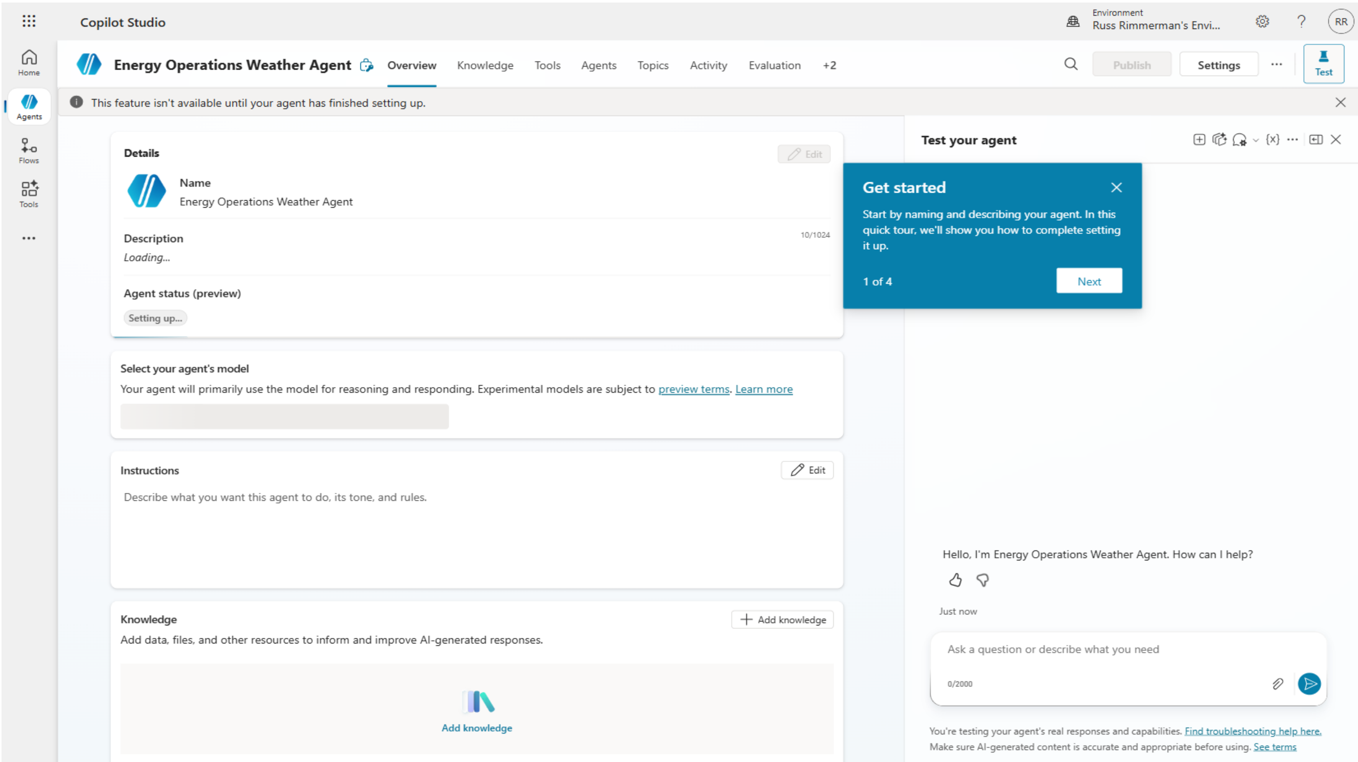
1.2 Configure Agent Identity

Step 3: In the “Name your agent” dialog, enter: Energy Operations Weather Agent. Add a description that explains its purpose for grid operations.



> **TIP:** *Choose descriptive names that clearly indicate the agent's domain. This helps with discoverability when you have multiple agents in your environment.*

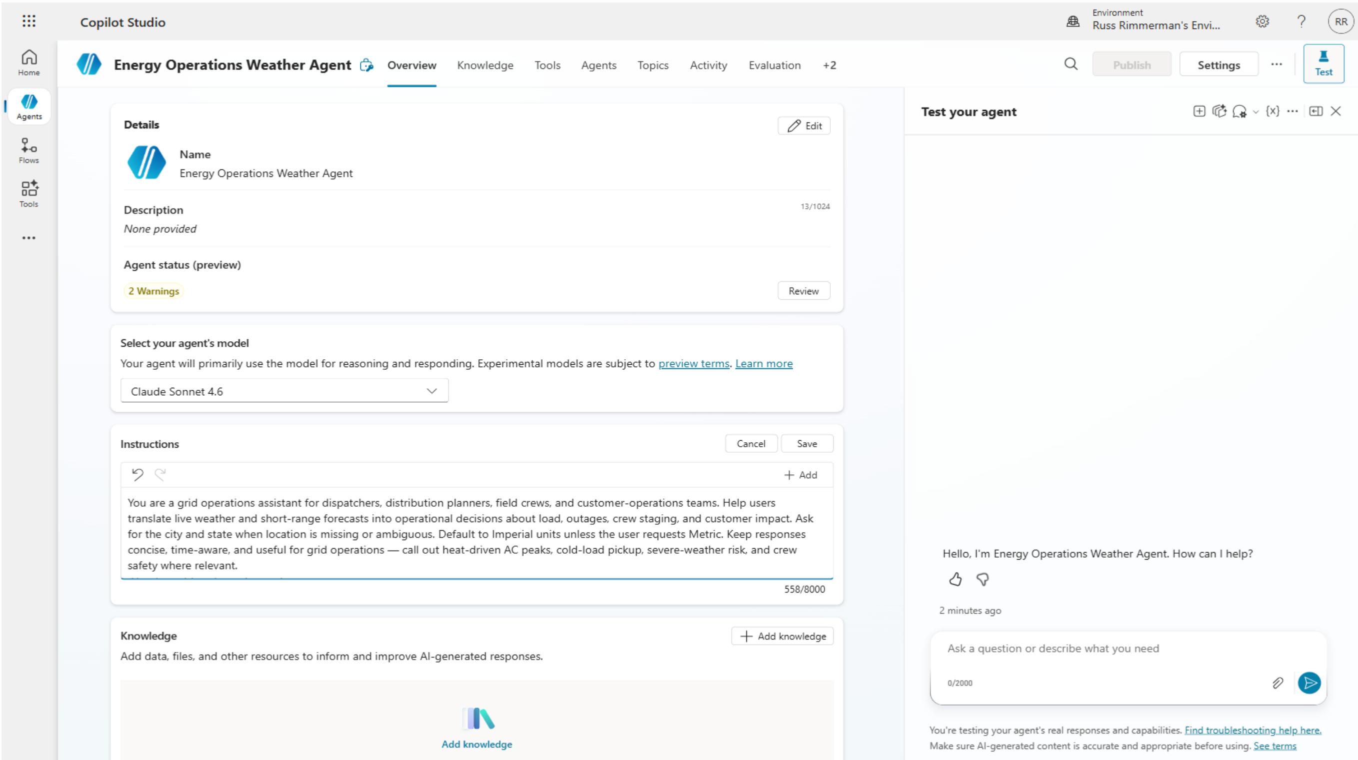
Step 4: After creation, you will see the Agent Overview page. This is your central hub for managing all aspects of the agent.



1.3 Add Agent Instructions

Instructions tell the AI model how to behave. For an energy operations agent, we need specific guidance about utility terminology, safety protocols, and response formatting.

Step 5: In the Instructions field, enter comprehensive guidance covering: role definition, weather analysis protocols, grid impact assessment criteria, and response formatting standards.



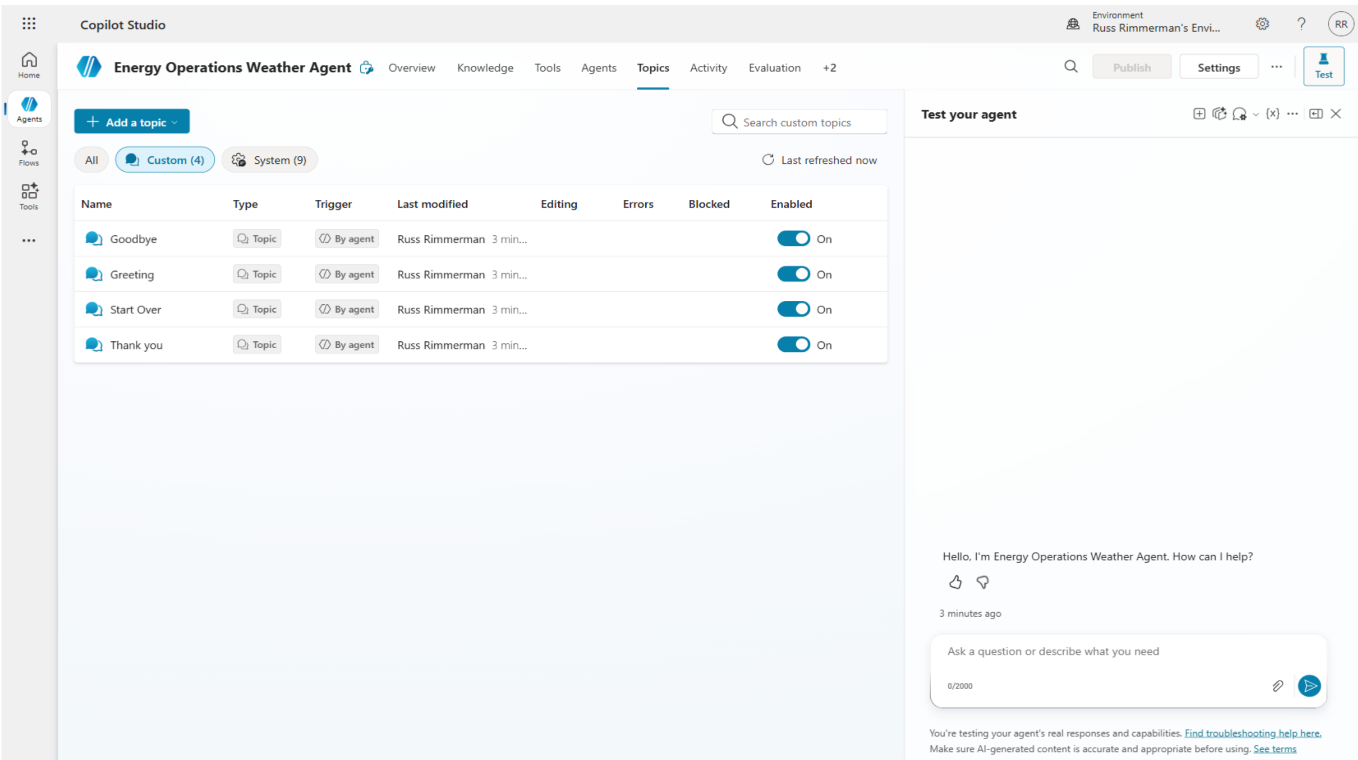
> ⚠ **NOTE:** *Instructions are critical for agent quality. They define the AI's persona, capabilities, and constraints. Well-written instructions prevent hallucination and ensure domain-appropriate responses.*

Section 2: Create a Topic with Variables

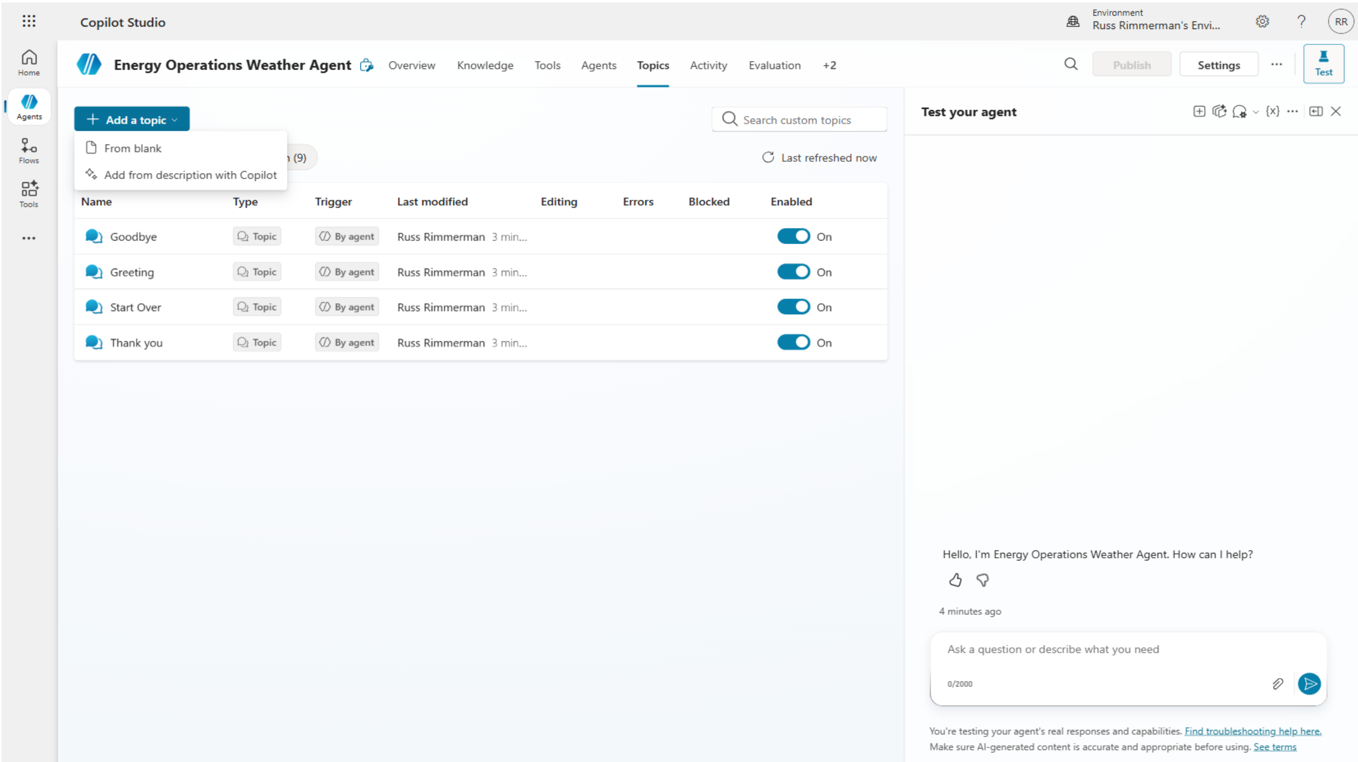
Topics are conversational flows that define how your agent handles specific intents. Variables store user inputs and intermediate data as the conversation progresses. In this section, you will create a “Service Territory Weather Lookup” topic that collects city and state information from the user.

2.1 Navigate to Topics

Step 1: Click the “Topics” tab in the agent navigation bar. This shows all topics (system and custom) for your agent.

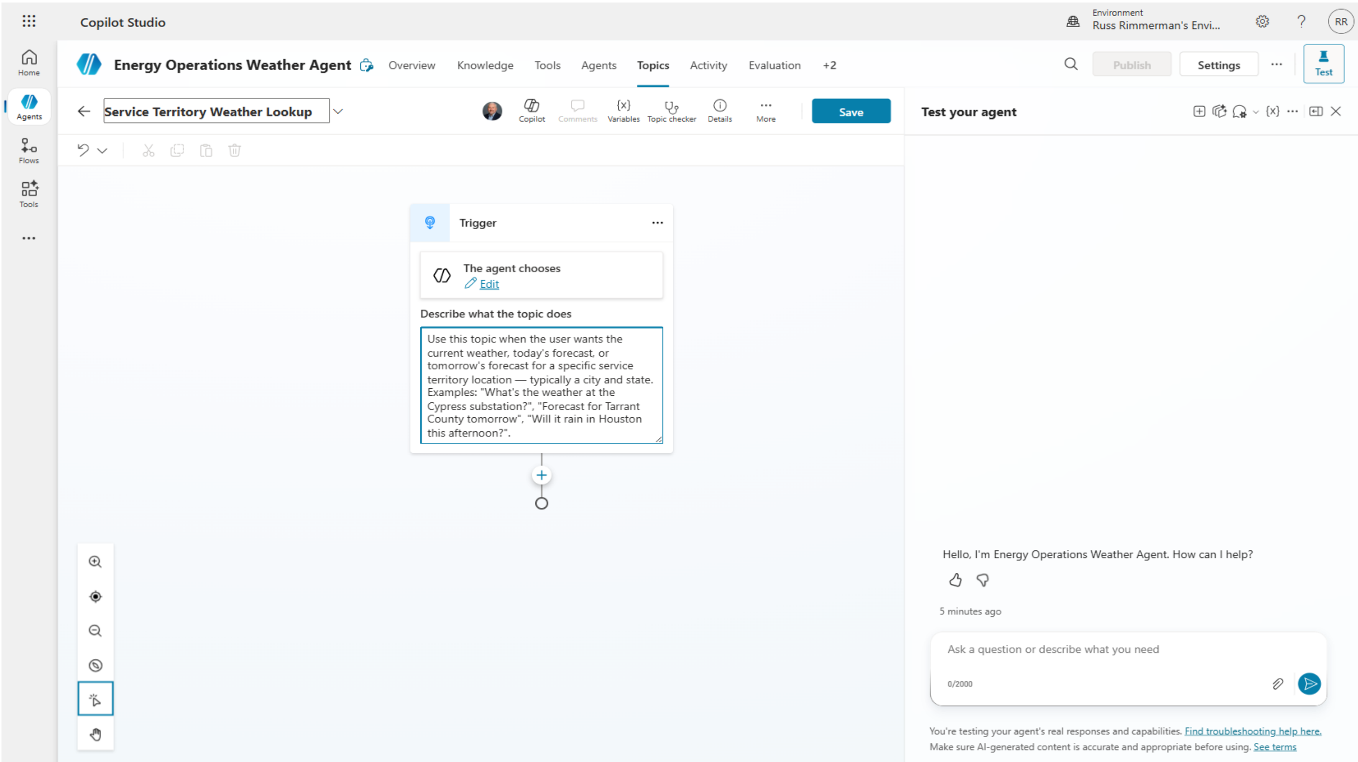


Step 2: Click “Add” → “From blank” to create a new custom topic.



2.2 Configure Topic Details

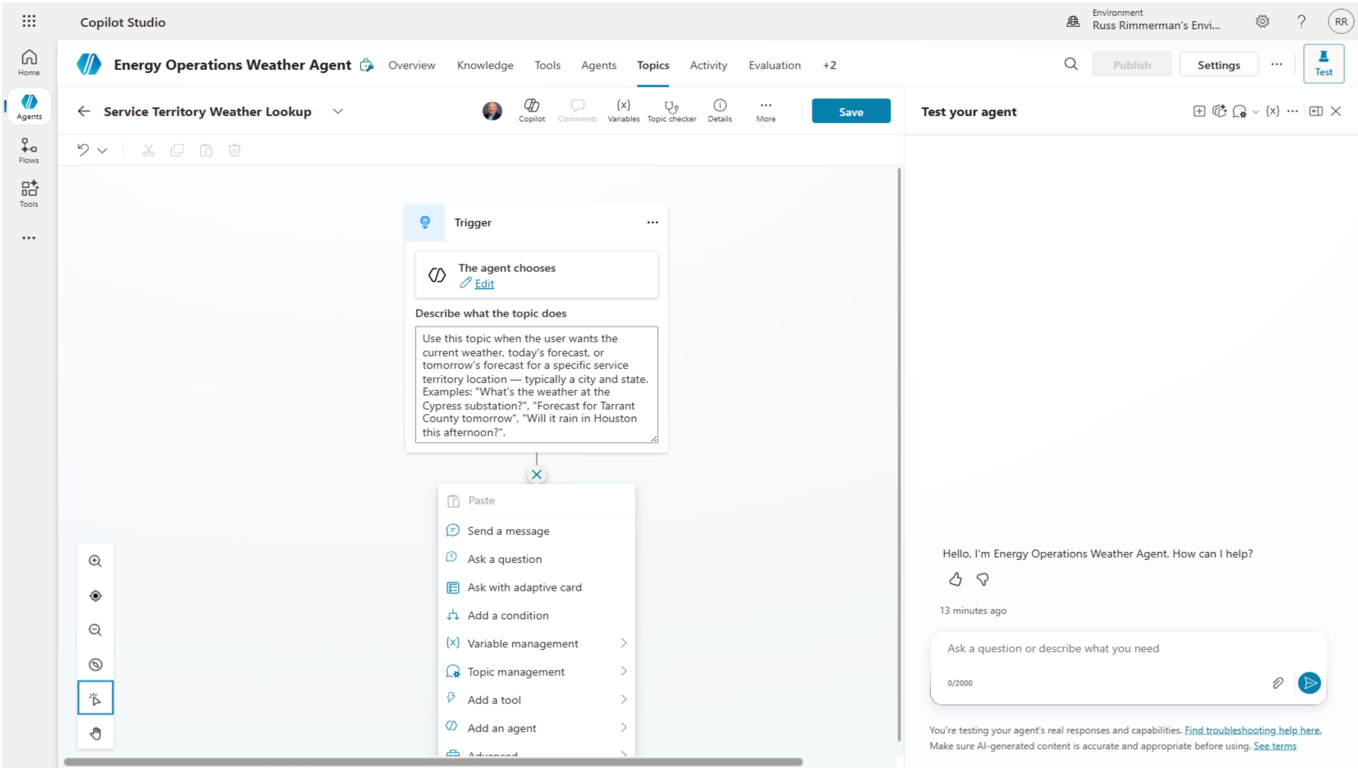
Step 3: Name the topic "Service Territory Weather Lookup" and add trigger phrases such as: "weather in", "current conditions", "forecast for", "grid weather".



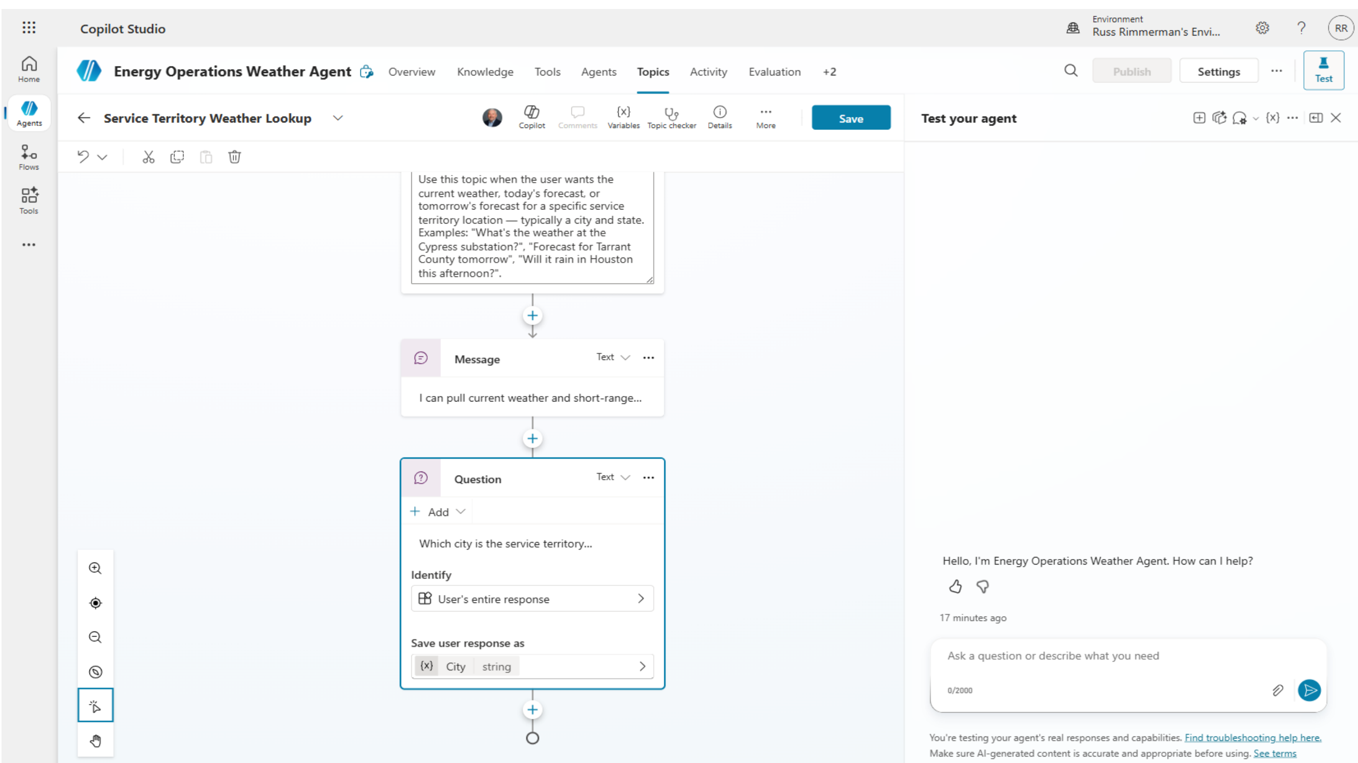
2.3 Add Question Nodes

Question nodes prompt the user for information and store their response in a variable. We need two pieces of data: city and state.

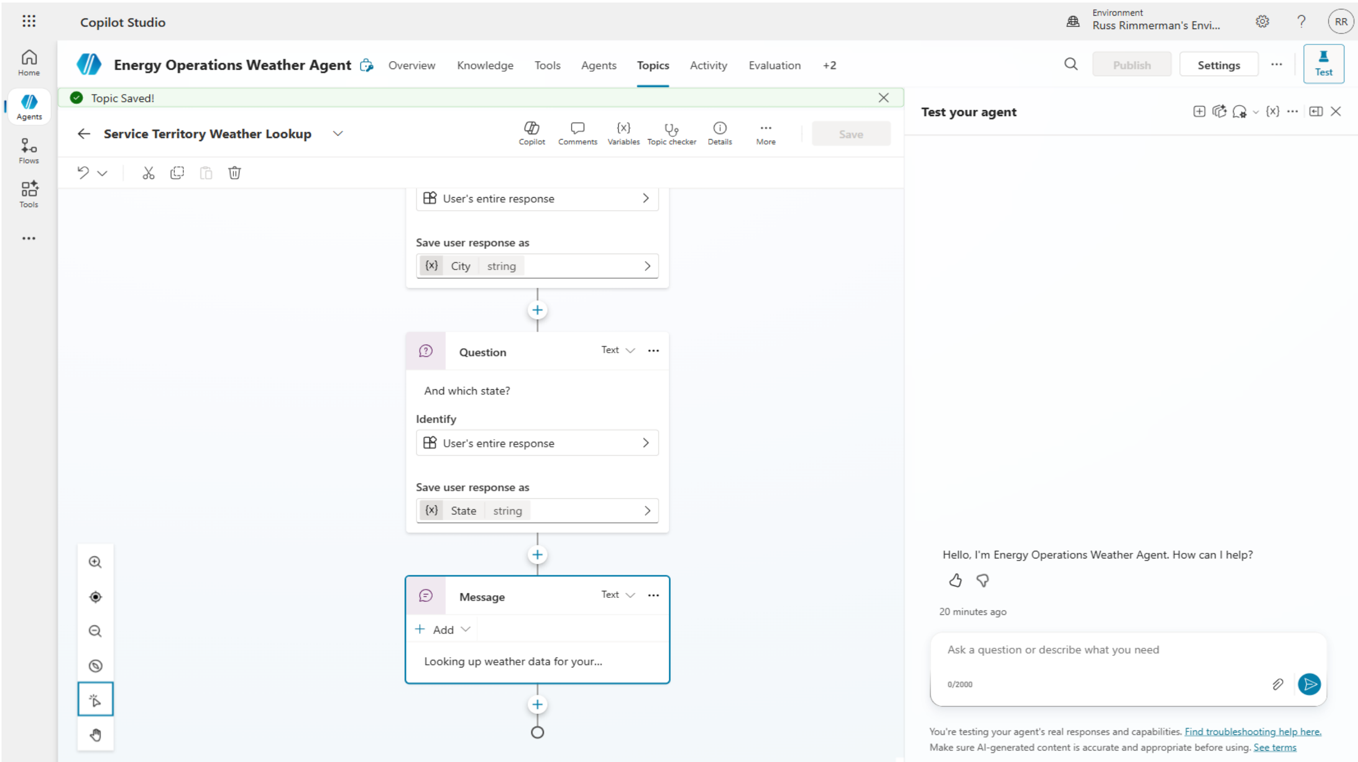
Step 4: Click the "+" button to add a node, then select "Ask a question."



Step 5: Configure the first question to ask "What city is the service territory located in?". Set the response to save to a variable named "City". Add a second question for State.



Step 6: Click "Save" to preserve your topic. You should see a "Topic Saved!" confirmation banner.



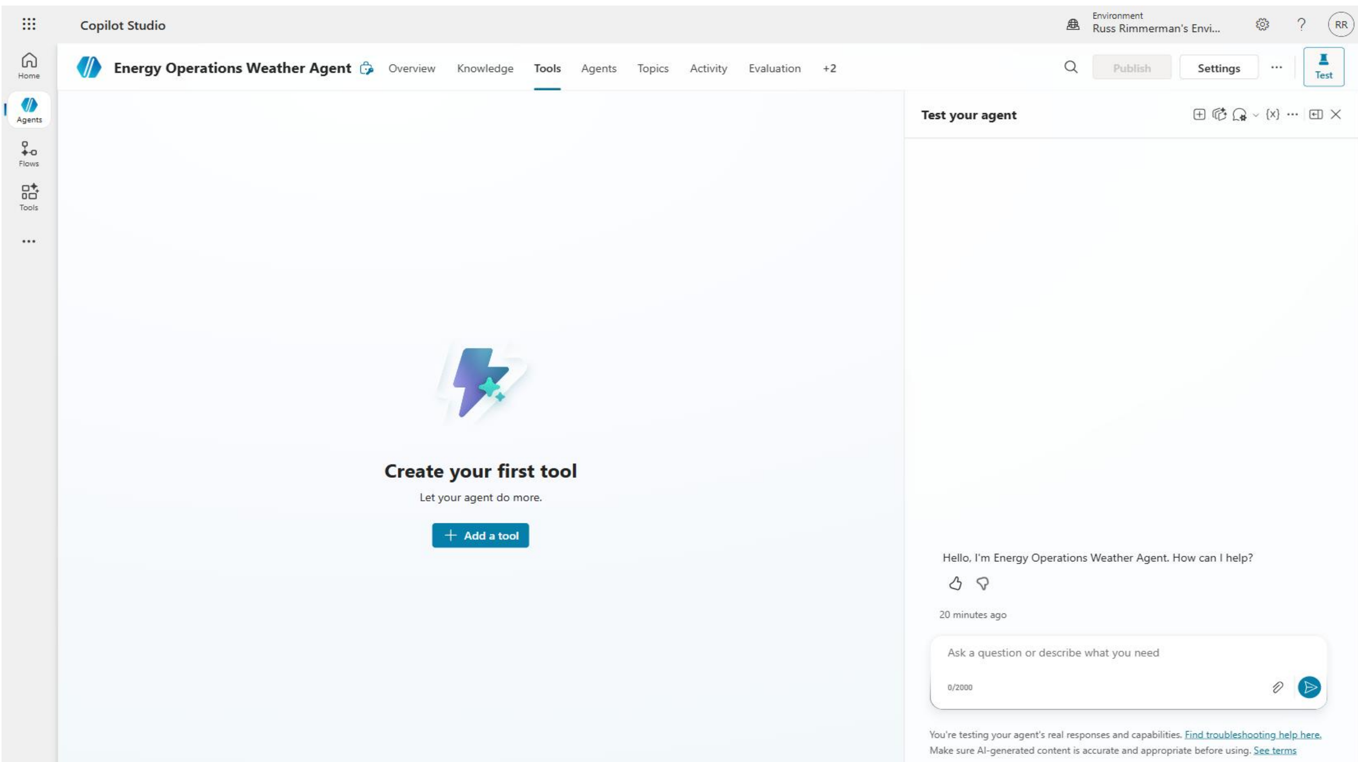
💡 **TIP:** *Always save your topic before navigating away. Unsaved changes will be lost.*

Section 3: Add MSN Weather Connector Tools

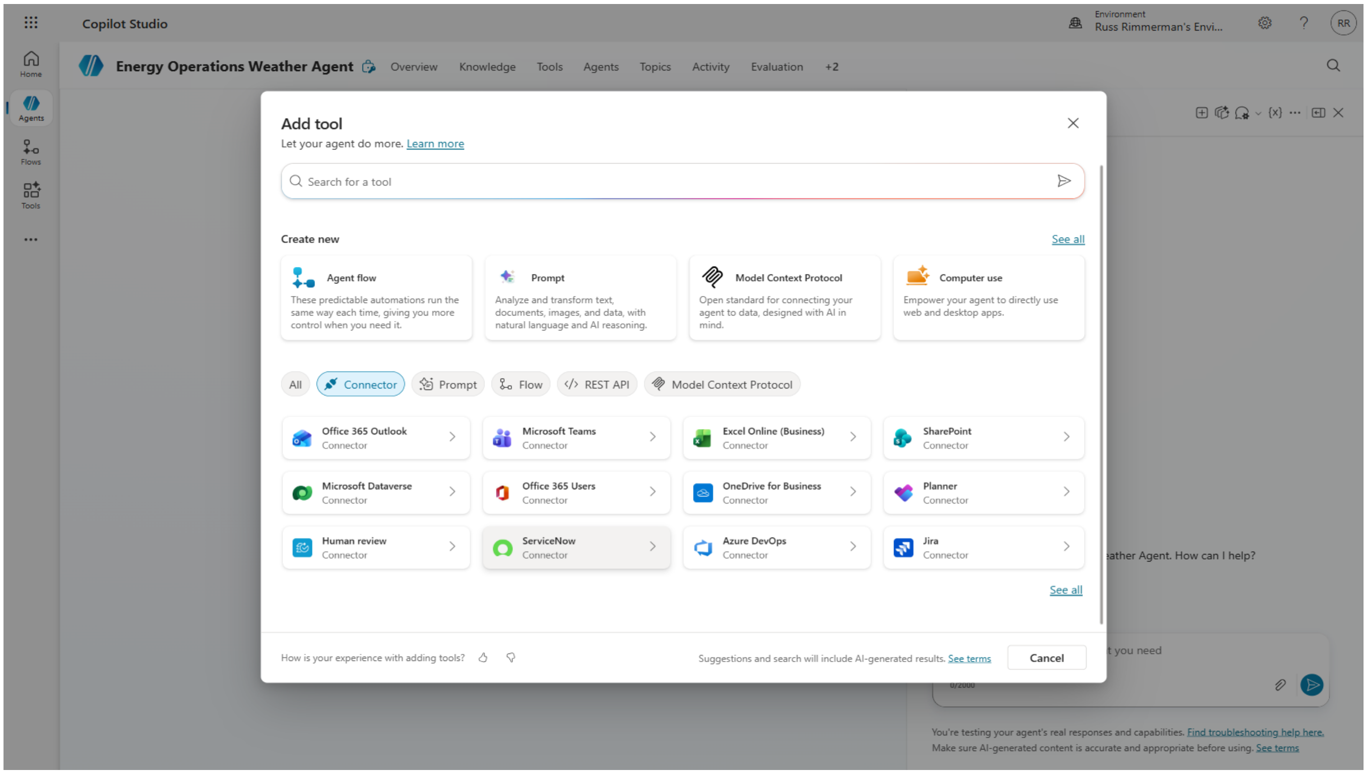
Tools connect your agent to external data sources and services. The MSN Weather connector provides real-time weather data including current conditions, today's forecast, and tomorrow's forecast. Adding multiple tools gives the agent flexibility to answer different types of weather questions.

3.1 Navigate to Tools

Step 1: Click the "Tools" tab. Initially, this will be empty since no tools have been added yet.

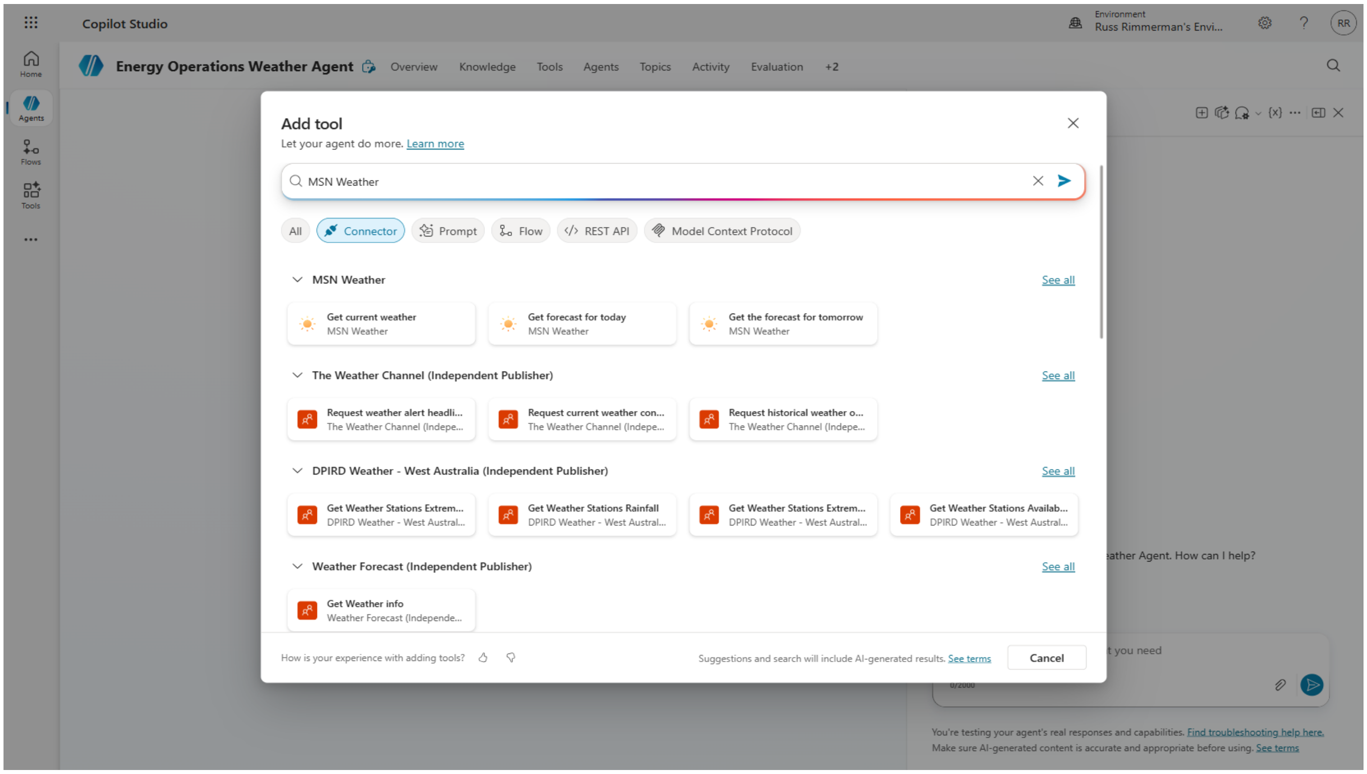


Step 2: Click "Add a tool" to open the tool gallery. This shows all available connectors, prompts, flows, and custom tools.

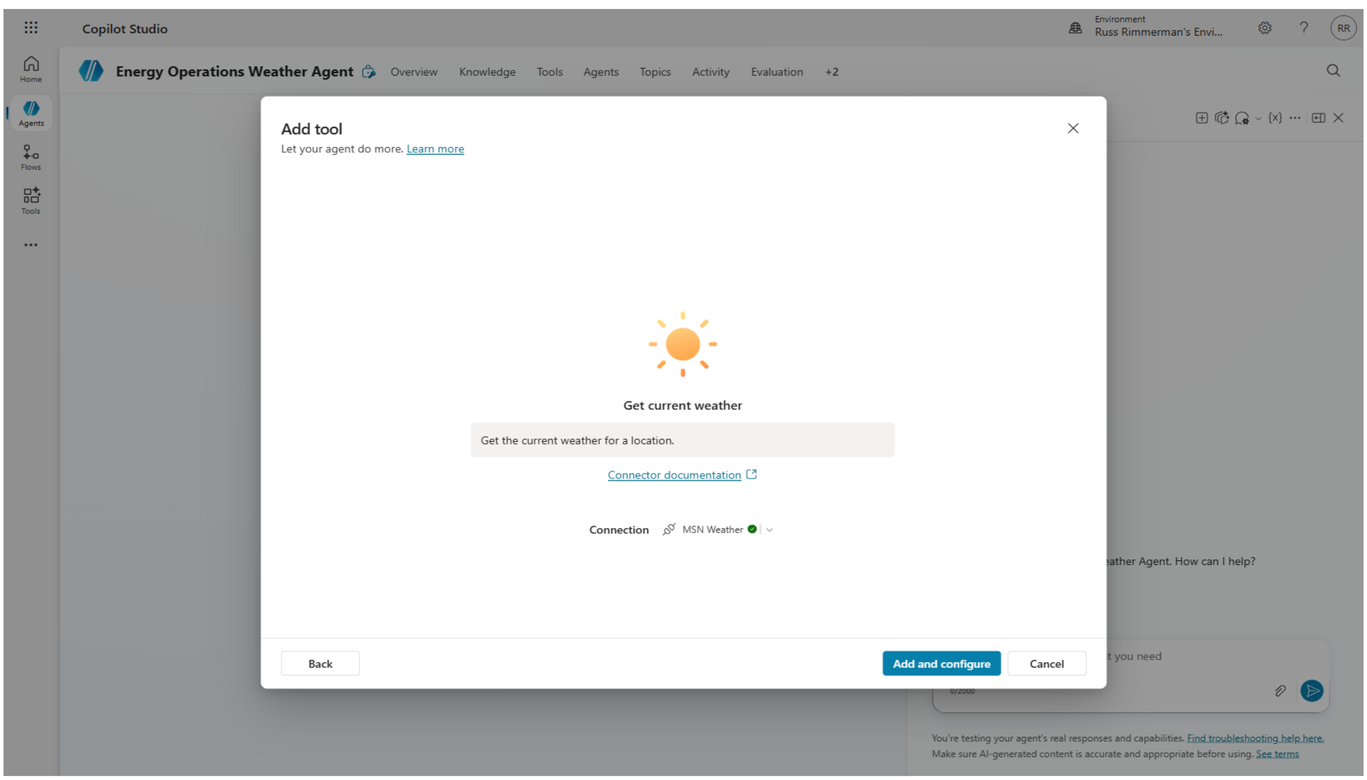


3.2 Add MSN Weather Connectors

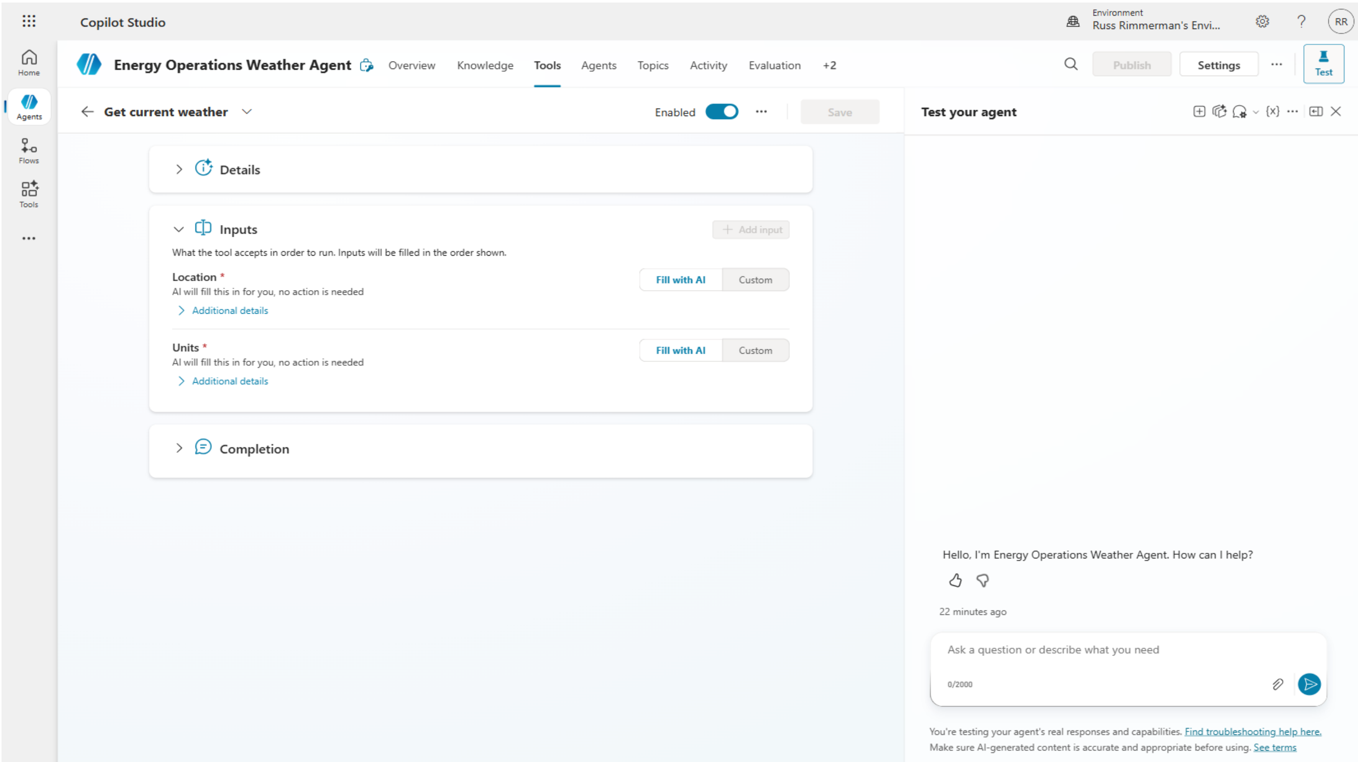
Step 3: Search for “MSN Weather” in the tool gallery. You will see three available actions: Get current weather, Get forecast for today, and Get the forecast for tomorrow.

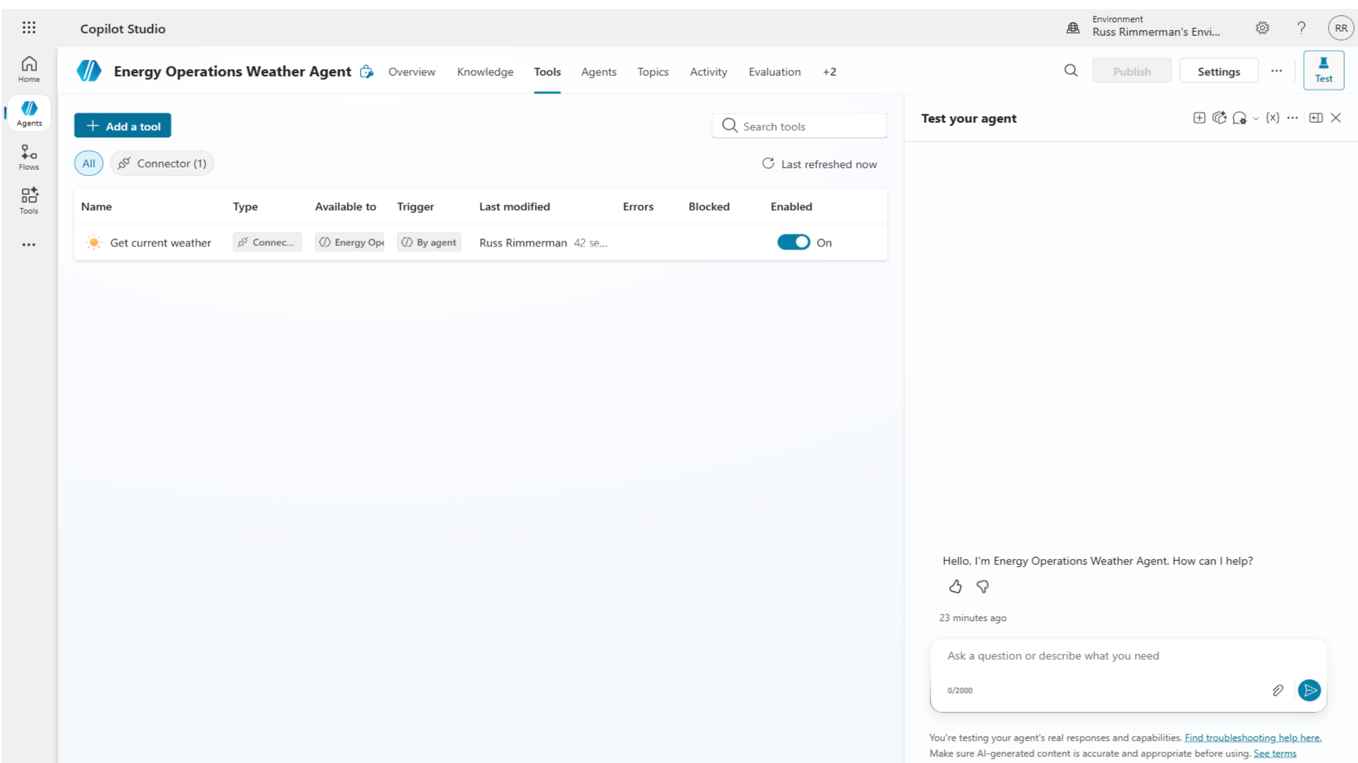


Step 4: Select "Get current weather" first. Review the tool details showing its inputs (Location, Units) and outputs (weather data object).

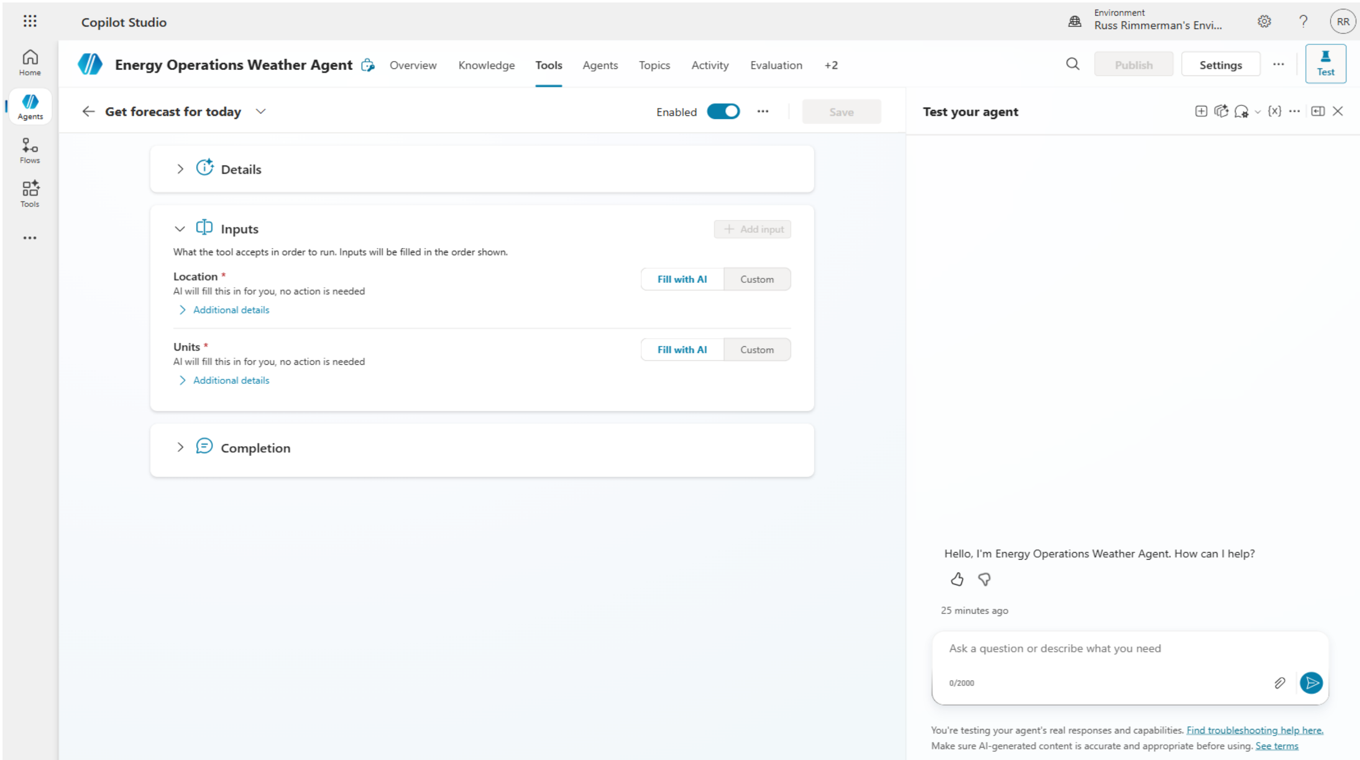


Step 5: Click "Add" to add the tool to your agent. Configure any required settings on the tool configuration page.

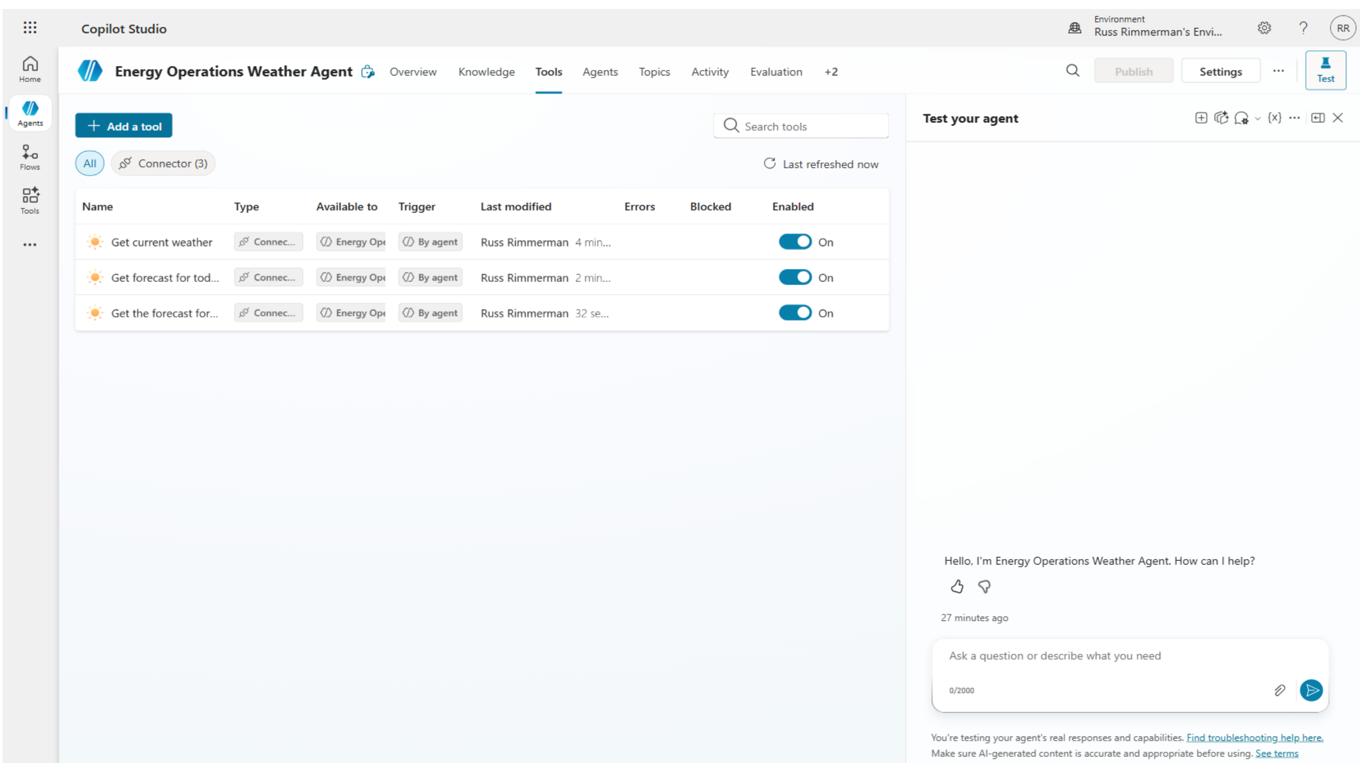




Step 6: Repeat the process for "Get forecast for today" and "Get the forecast for tomorrow."



Step 7: Verify all three tools appear in your Tools list.



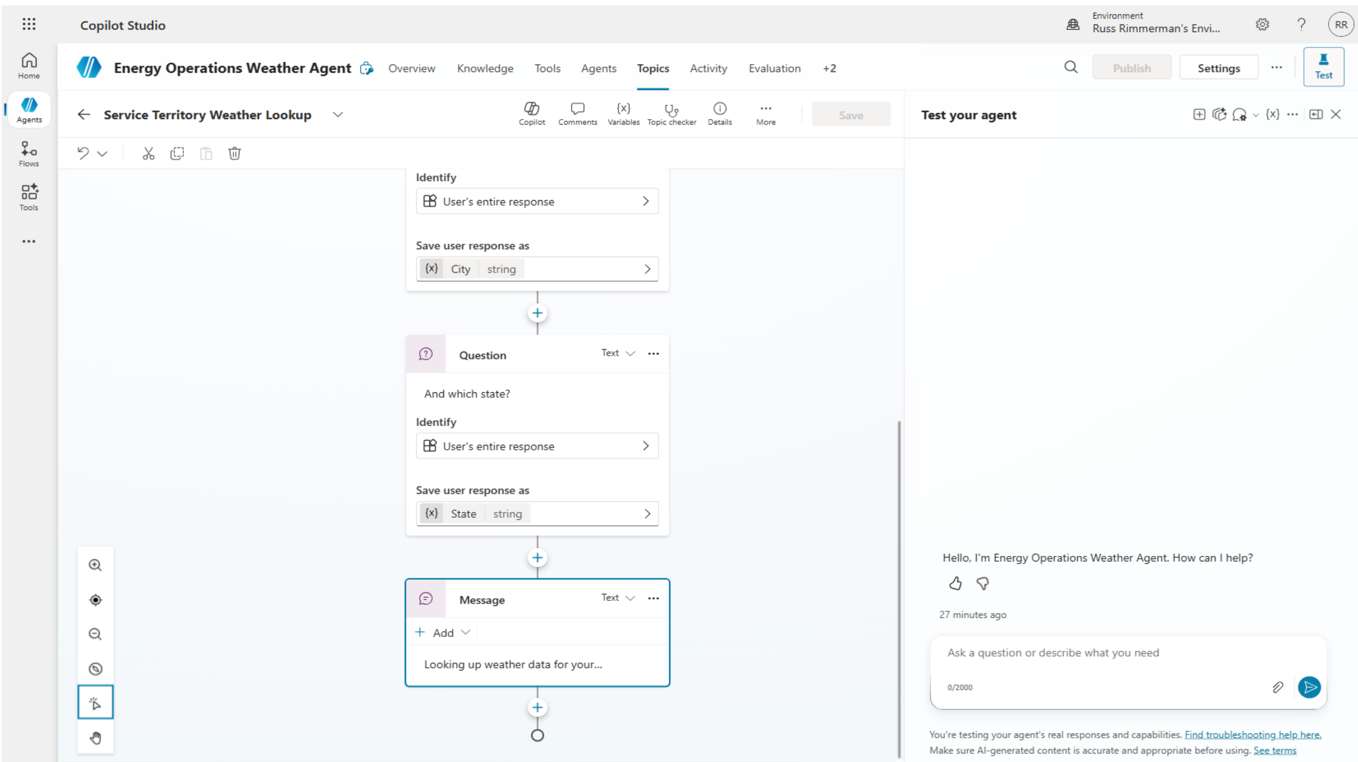
> ⚠ **NOTE:** *Adding all three weather tools gives the orchestrator options. It can intelligently select which tool to call based on the user's question (current vs. forecast).*

Section 4: Wire Tools into Topic with Power Fx

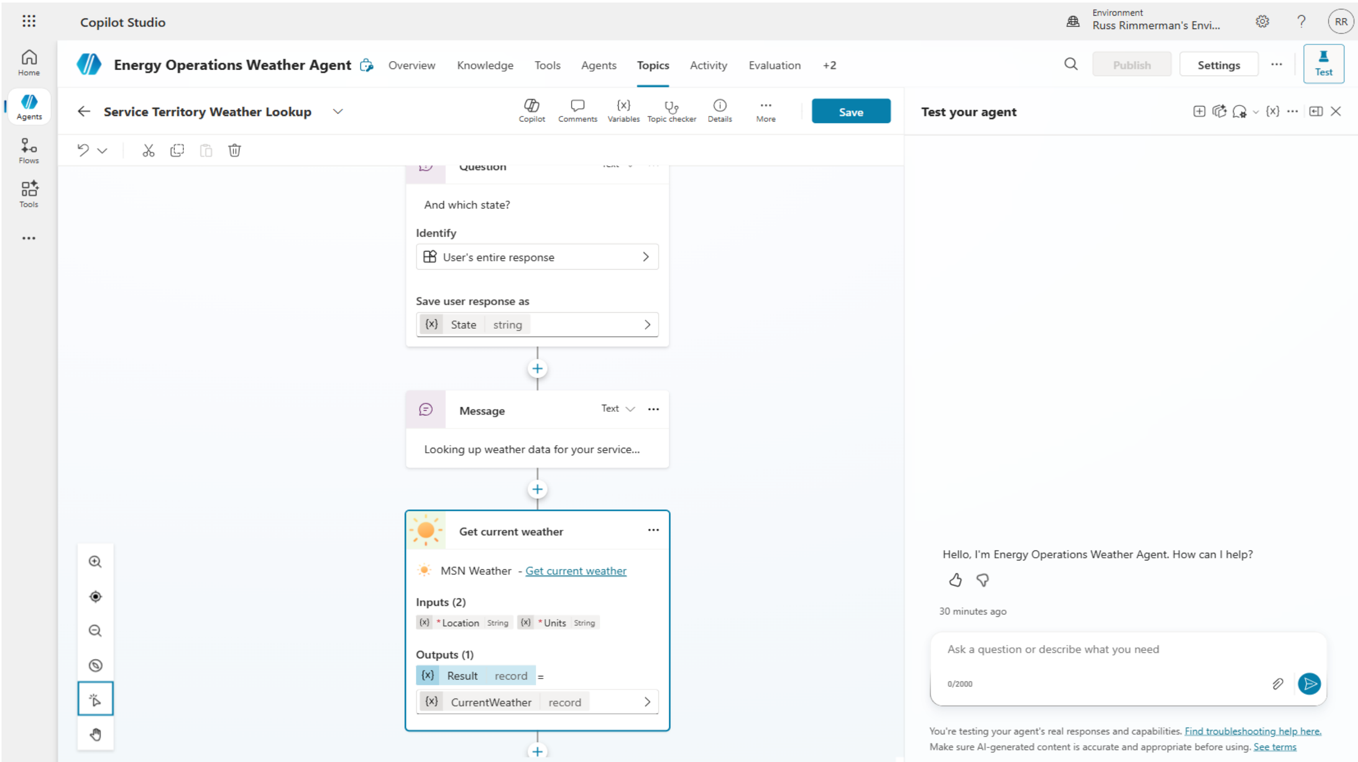
Power Fx is Microsoft's low-code formula language (similar to Excel). In Copilot Studio, Power Fx connects your topic variables to tool inputs and formats tool outputs for display. This is where your conversational flow becomes a functioning data pipeline.

4.1 Add Tool Node to Topic

Step 1: Return to the "Service Territory Weather Lookup" topic. Your current flow has the two question nodes. Click "+" after the State question to add a new node.



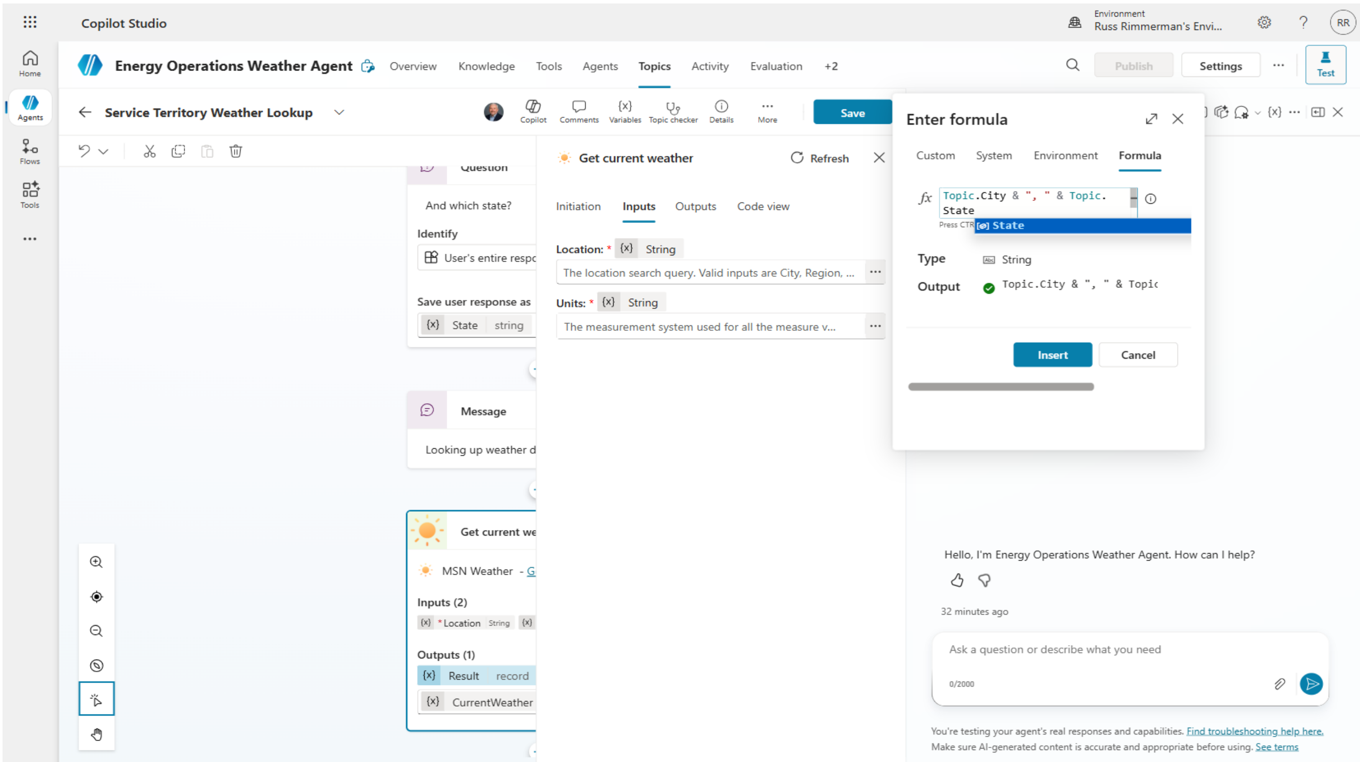
Step 2: Select "Call an action" and choose the "Get current weather" tool from the list.



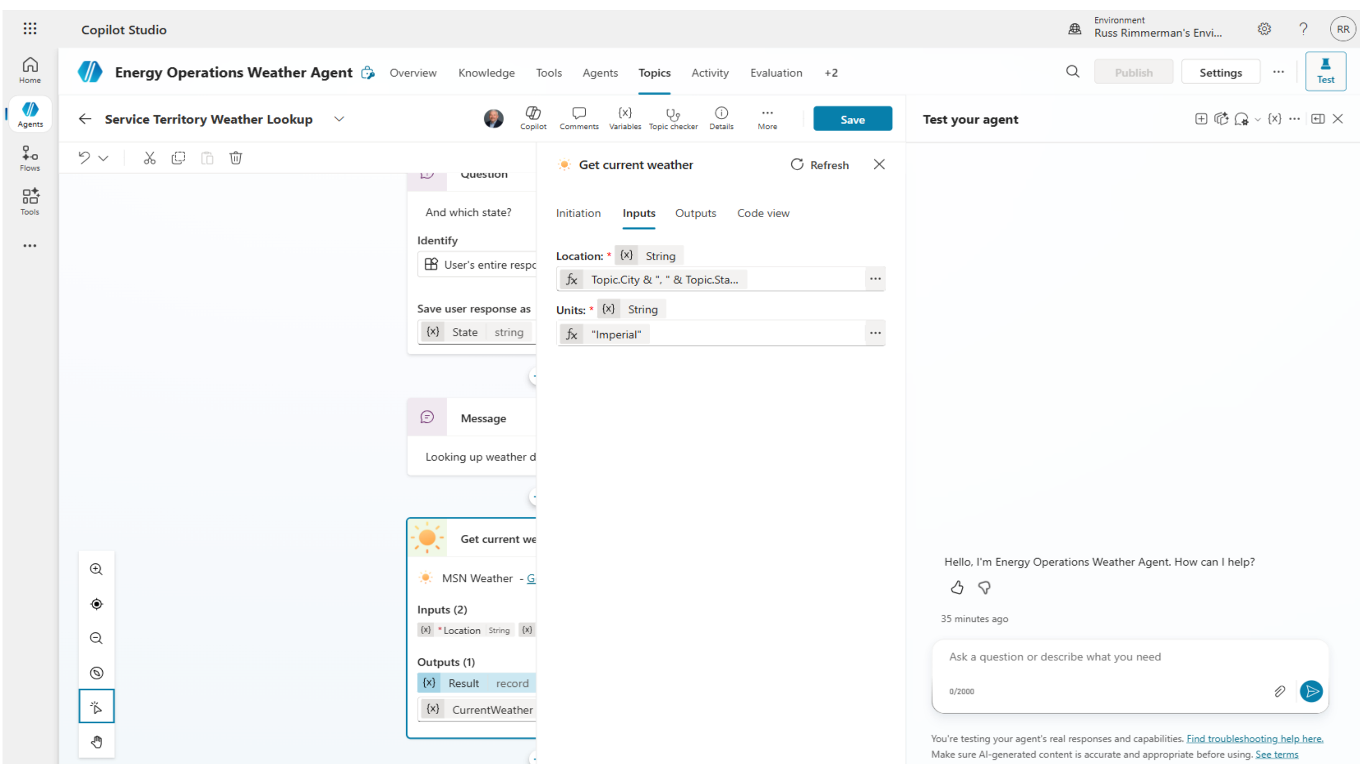
4.2 Configure Power Fx Input Formulas

The weather tool requires a Location input (city, state format) and Units (Imperial/Metric). We use Power Fx to combine our topic variables into the correct format.

Step 3: Click the Location input field and enter the Power Fx formula: Topic.City & ", " & Topic.State



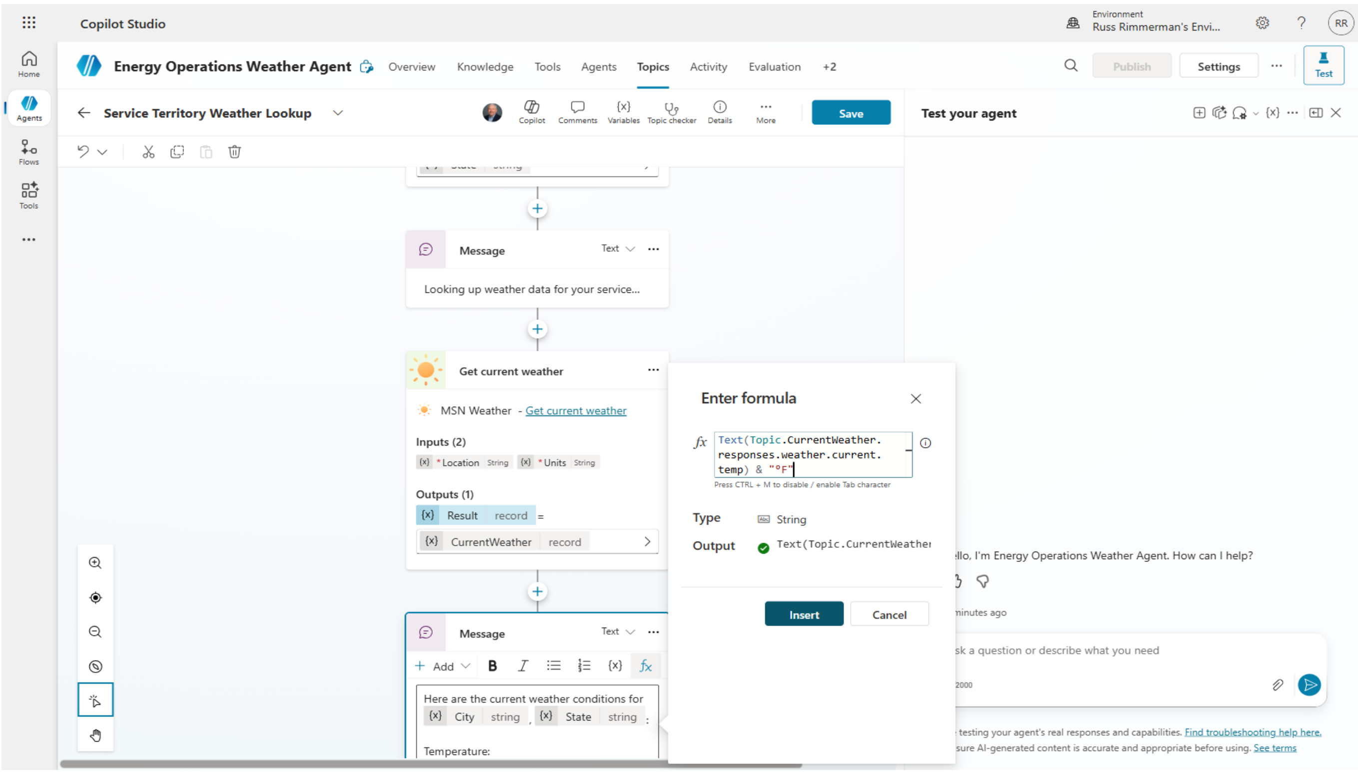
Step 4: Set the Units input to the formula: "Imperial" (hardcoded for US energy operations).



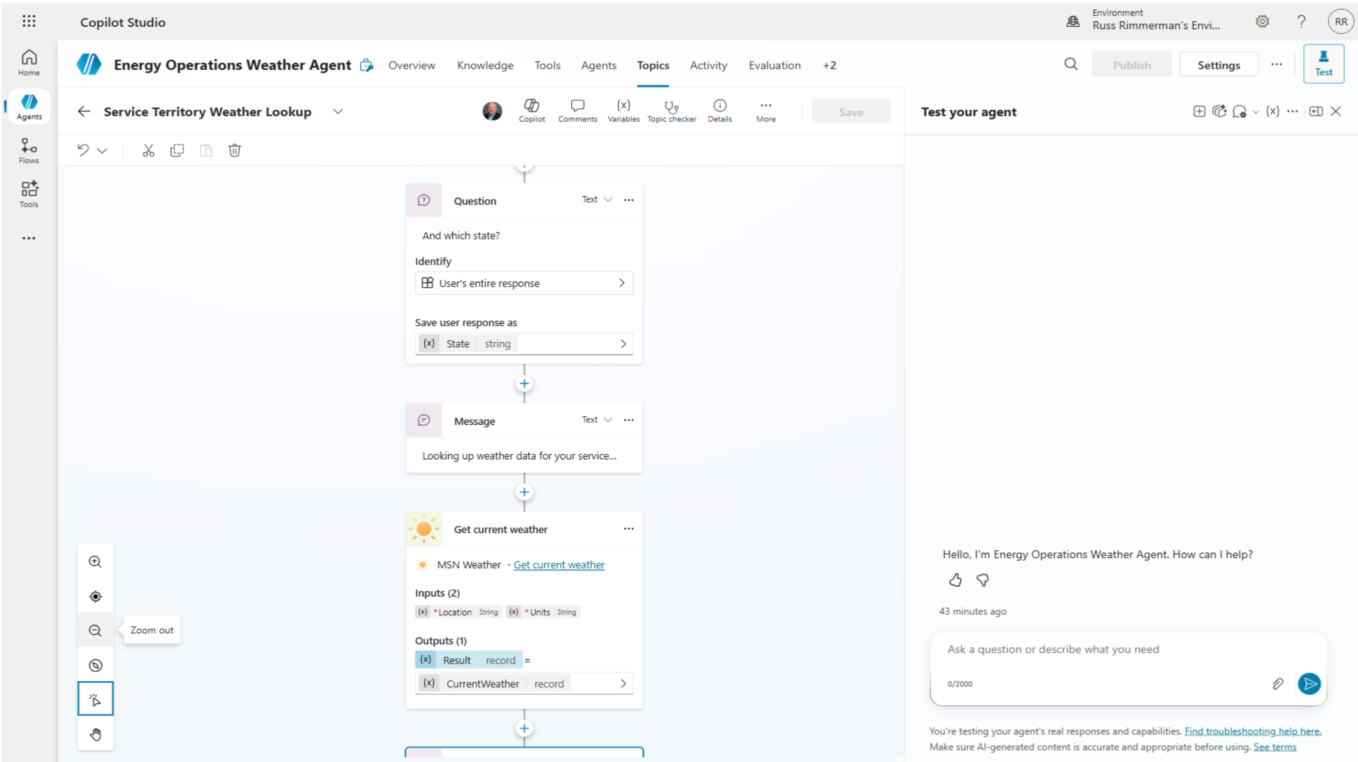
4.3 Add Response Message

After the tool executes, we need to display results to the user. The tool returns a complex object with nested weather properties.

Step 5: Add a Message node after the tool node. Use Power Fx expressions to format the response, referencing properties like Topic.CurrentWeather.responses.weather.current.temp.



Step 6: Review the complete topic flow showing: Questions → Tool Call → Response Message.



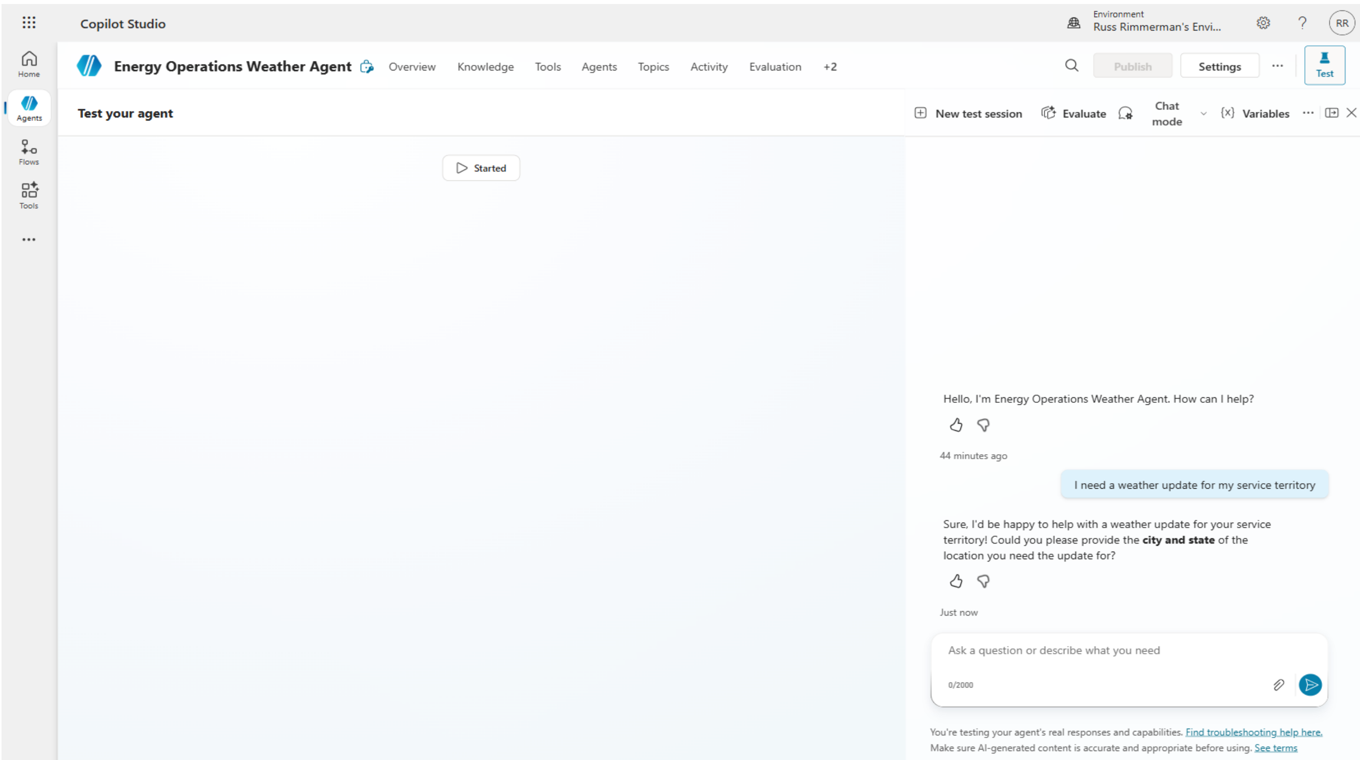
> **TIP:** *Power Fx formulas are validated in real-time. A green checkmark means the formula is syntactically correct. Red errors indicate type mismatches or invalid property references.*

Section 5: Test the Agent

Testing validates that your entire flow works end-to-end: the agent triggers the correct topic, collects user inputs, calls the weather API with the right parameters, and returns formatted results. The built-in Test panel simulates a real conversation.

5.1 Open Test Panel

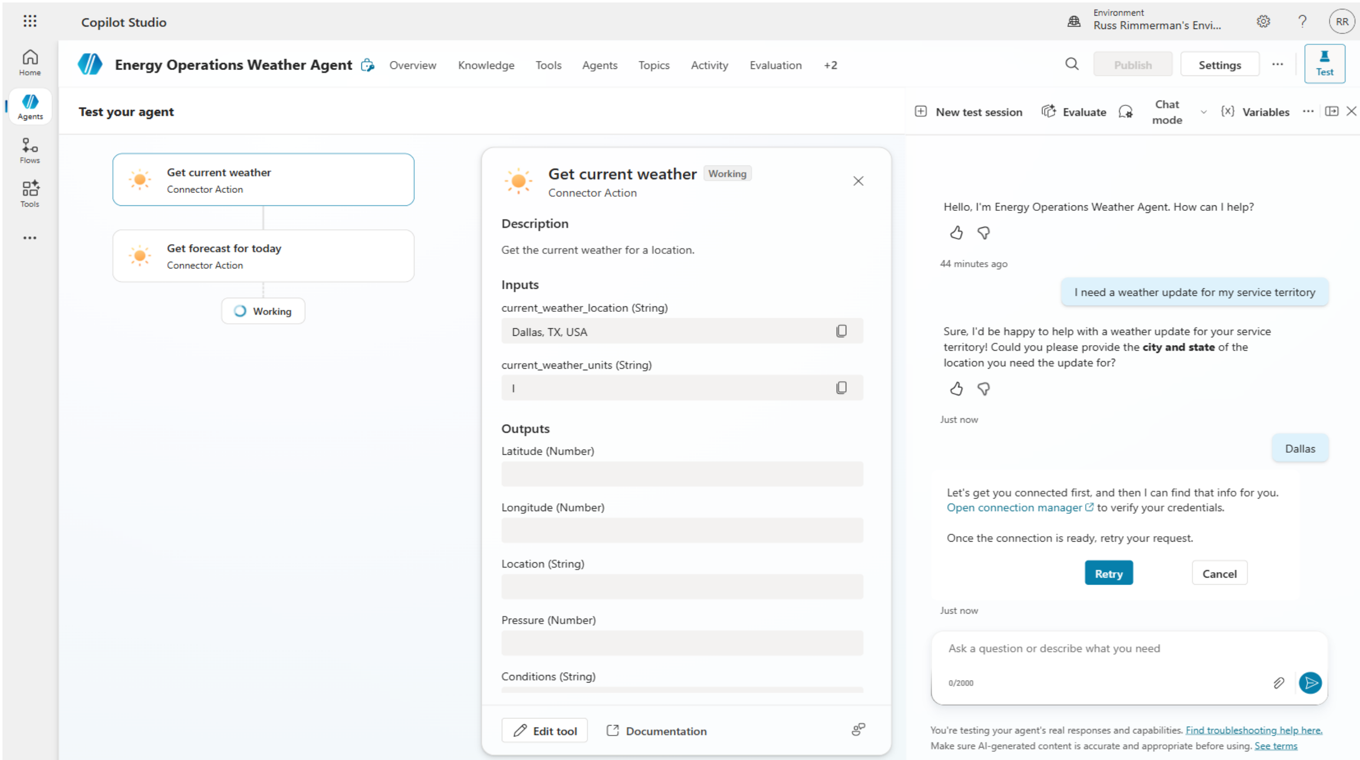
Step 1: Click the “Test” button in the upper right corner to open the test chat panel.

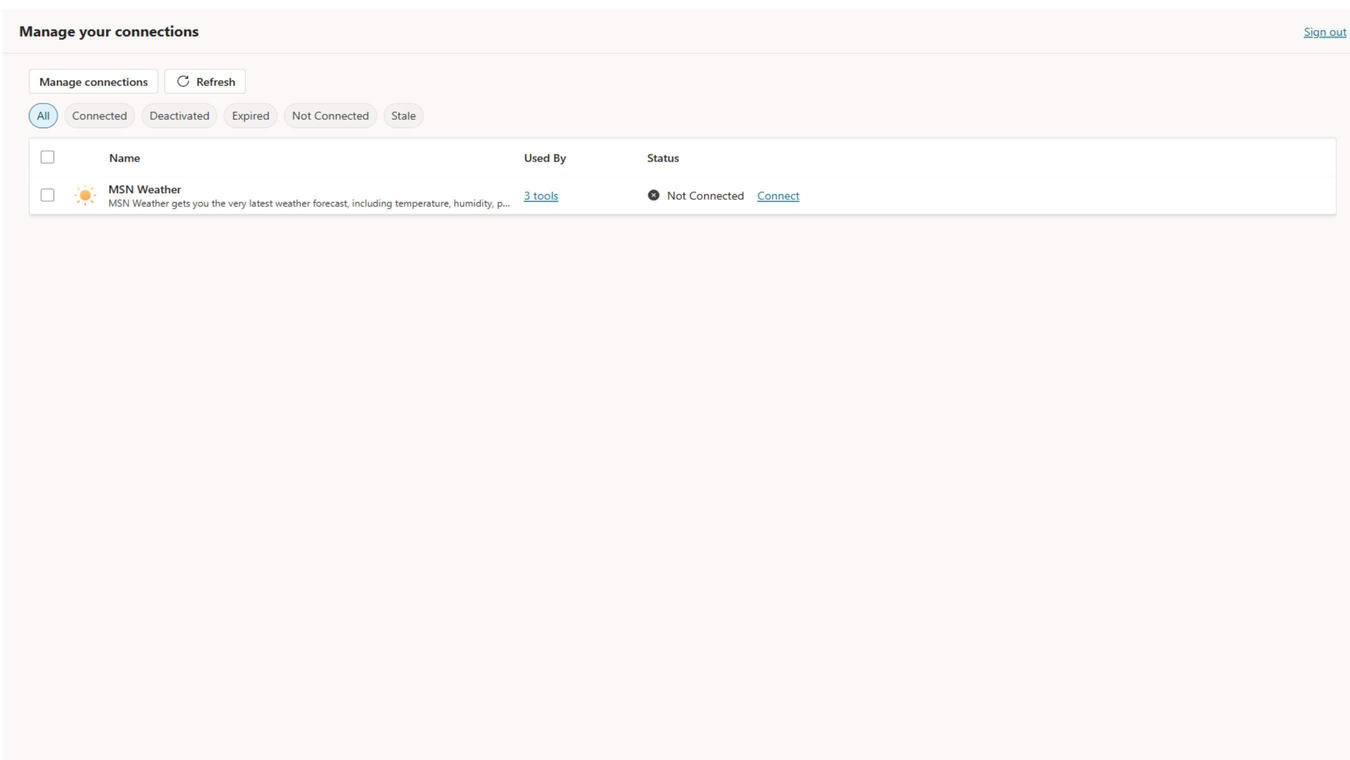


5.2 Authenticate Connector

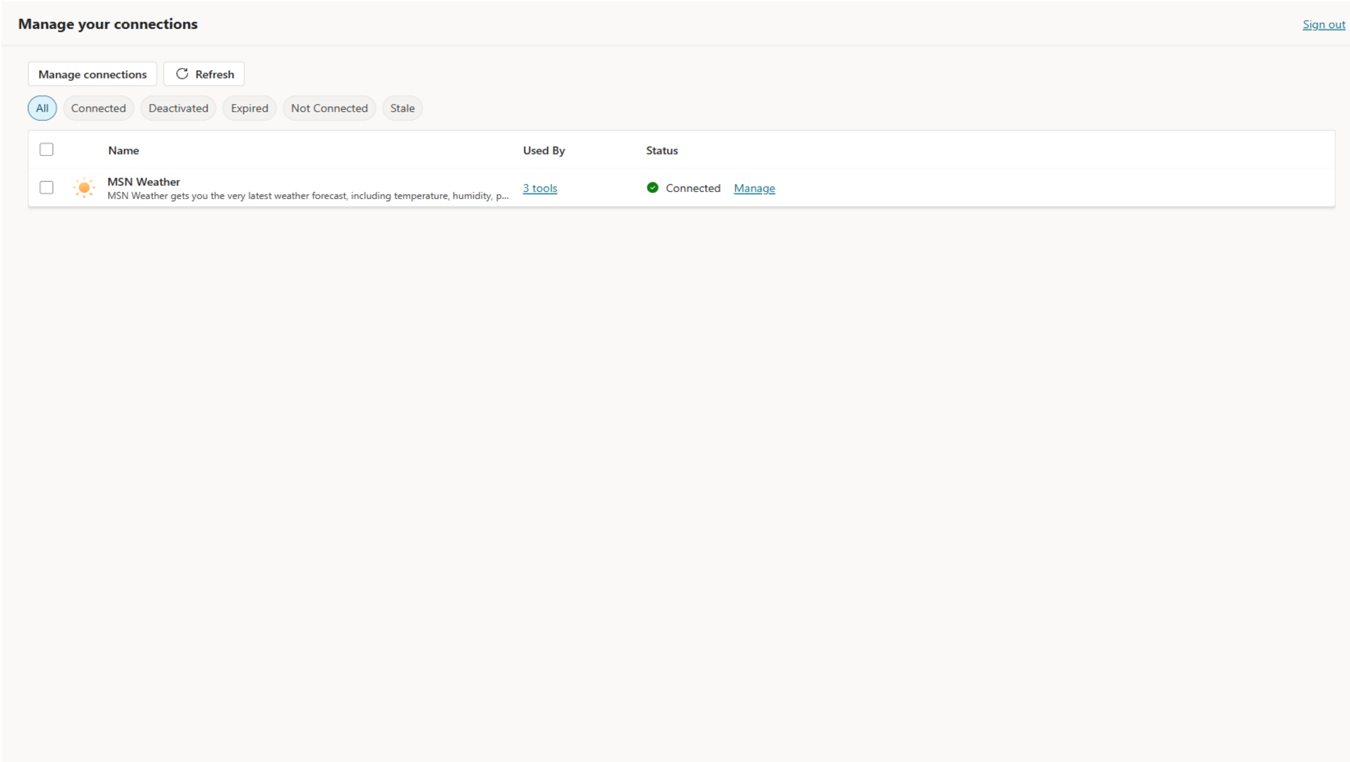
The first time a connector is used, it requires authentication. This is a one-time step per connector per user.

Step 2: When prompted to connect, click the connection link to authenticate with MSN Weather.



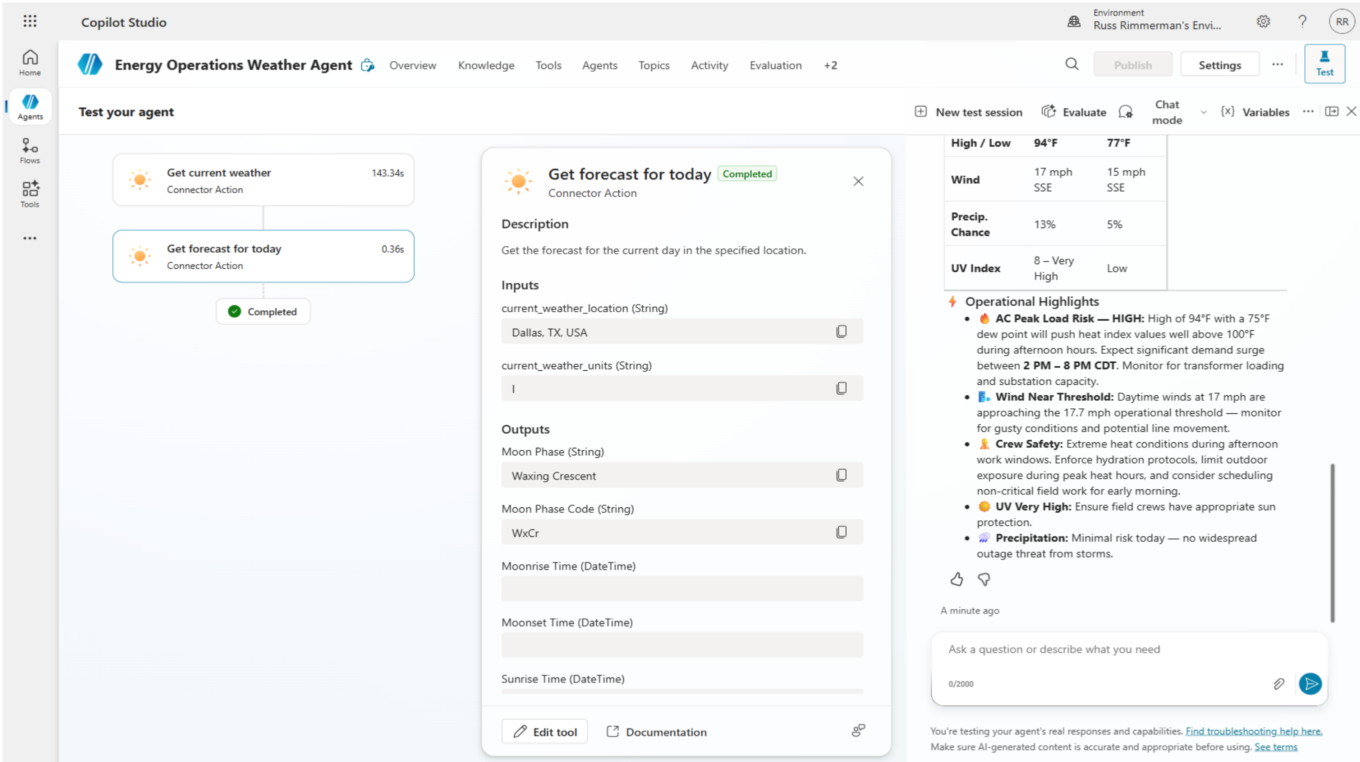


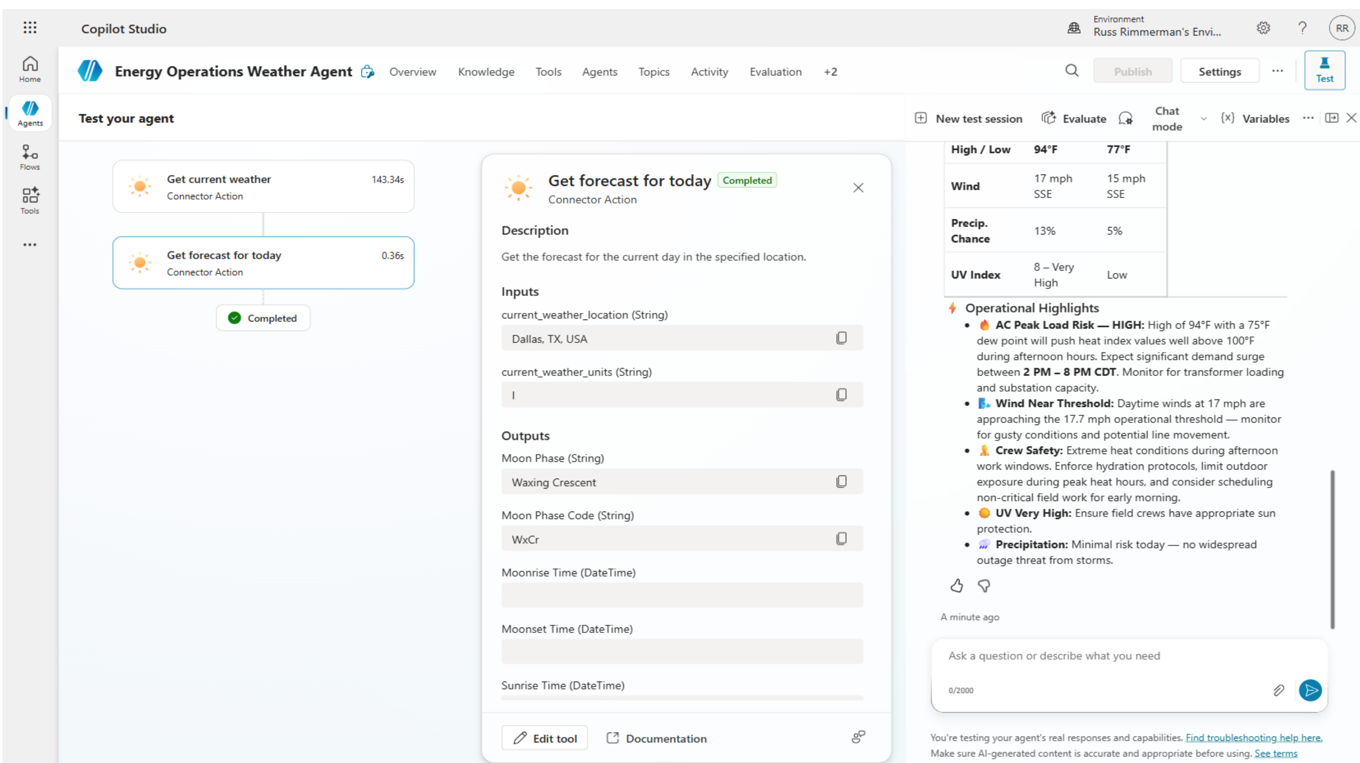
Step 3: After authenticating, verify the connection shows as established.

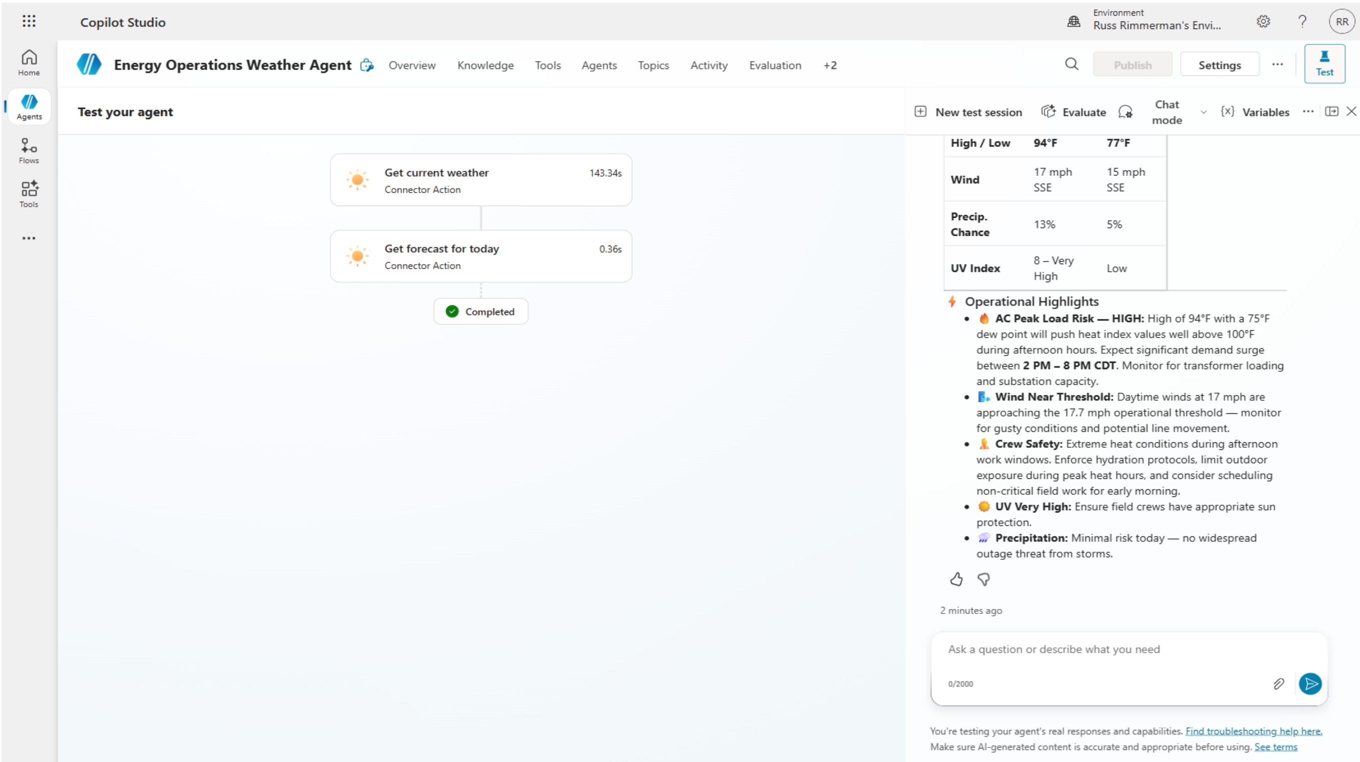


5.3 End-to-End Test

Step 4: Type a weather query such as "What's the weather in Dallas, Texas?" The agent should ask for city/state (or extract from your message), call the weather API, and return live conditions.


Copilot Studio
Energy Operations Weather Agent
Test your agent
Get current weather
Connector Action
143.34s
Get forecast for today
Connector Action
0.36s
Completed
Get forecast for today
Completed
Description
Get the forecast for the current day in the specified location.
Inputs
current_weather_location (String)
Dallas, TX, USA
current_weather_units (String)
I
Outputs
Moon Phase (String)
Waxing Crescent
Moon Phase Code (String)
WxCr
Moonrise Time (DateTime)
Moonset Time (DateTime)
Sunrise Time (DateTime)
Edit tool
Documentation
High / Low 94°F 77°F
Wind 17 mph SSE 15 mph SSE
Precip. Chance 13% 5%
UV Index 8 – Very High Low
Operational Highlights
AC Peak Load Risk — HIGH: High of 94°F with a 75°F dew point will push heat index values well above 100°F during afternoon hours. Expect significant demand surge between 2 PM – 8 PM CDT. Monitor for transformer loading and substation capacity.
Wind Near Threshold: Daytime winds at 17 mph are approaching the 17.7 mph operational threshold — monitor for gusty conditions and potential line movement.
Crew Safety: Extreme heat conditions during afternoon work windows. Enforce hydration protocols, limit outdoor exposure during peak heat hours, and consider scheduling non-critical field work for early morning.
UV Very High: Ensure field crews have appropriate sun protection.
Precipitation: Minimal risk today — no widespread outage threat from storms.
A minute ago
Ask a question or describe what you need
You're testing your agent's real responses and capabilities. Find troubleshooting help here.
Make sure AI-generated content is accurate and appropriate before using. See terms





> ⚠ **NOTE:** *Live weather data means your results will differ from these screenshots. That's expected! Verify the structure (temperature, conditions, humidity, wind) is correct, not the specific values.*

Section 6: Create a Custom Prompt Tool

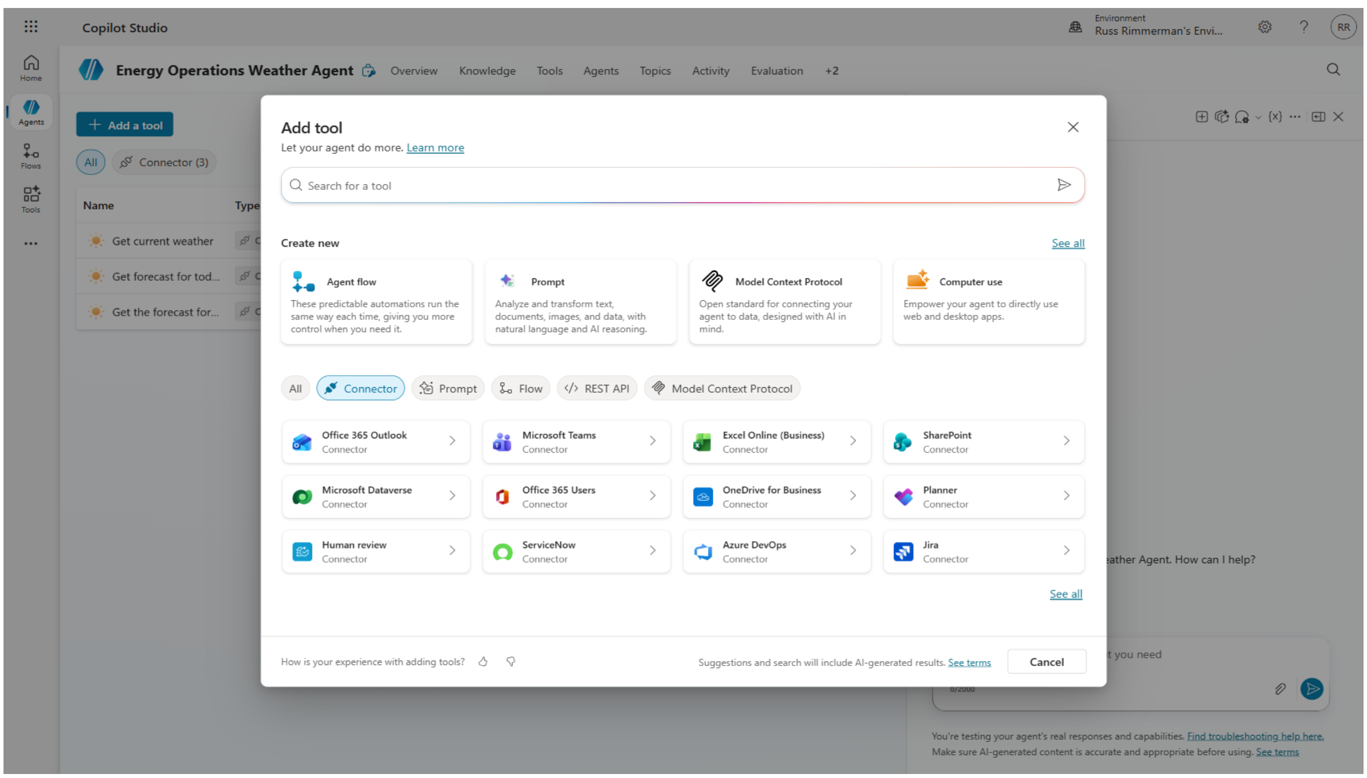
Custom Prompt Tools use generative AI to transform data into insights. Instead of just showing raw weather data, we will create a prompt that acts as an "Energy Operations Analyst" — taking weather data and generating a structured operations briefing with grid impact analysis, demand forecasts, and recommended actions.

Why this matters:

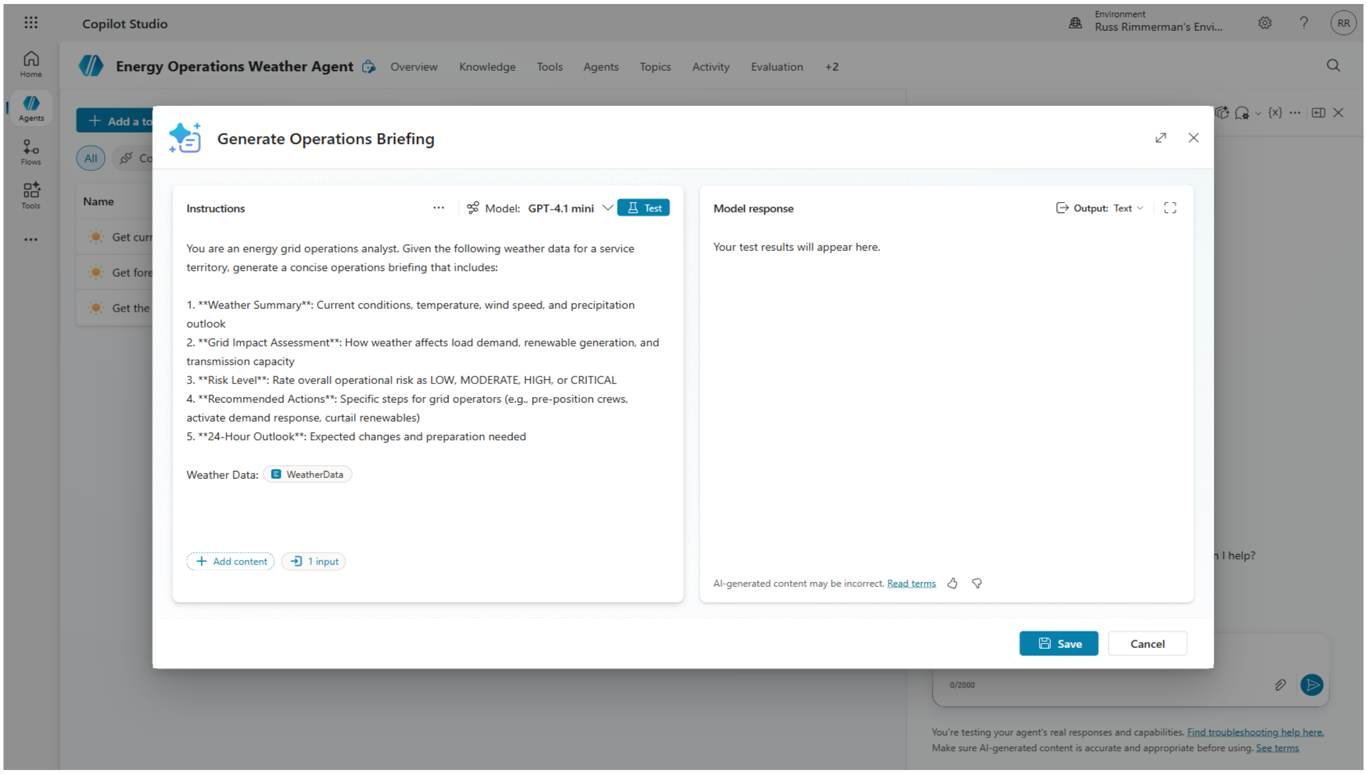
Raw data requires human interpretation. A prompt tool automates the analysis step, turning a temperature reading into an actionable "peak demand expected, pre-position crews" recommendation.

6.1 Create the Prompt

Step 1: Navigate to Tools → "Add a tool" and select "Prompt" from the "Create new" section.



Step 2: Name the prompt "Generate Operations Briefing" and enter detailed instructions that define the AI's role as an energy operations analyst. Include sections for: Grid Operations Implications, Active Weather Alerts analysis, Demand Forecast, and Recommended Actions.



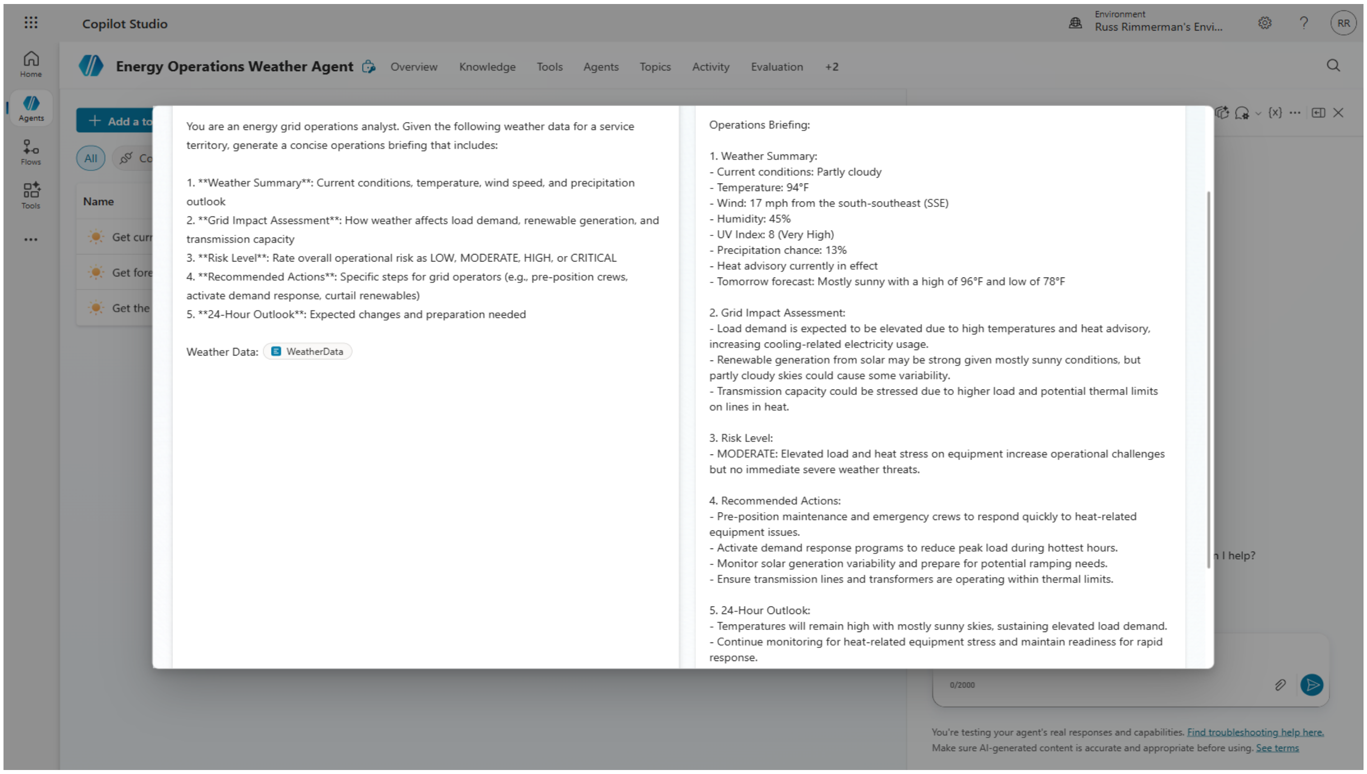
6.2 Add Input Variable

The prompt needs weather data to analyze. We create an input variable that will receive the formatted weather data from the connector output.

Step 3: Add a text input variable called "WeatherData" with sample data for testing the prompt independently.

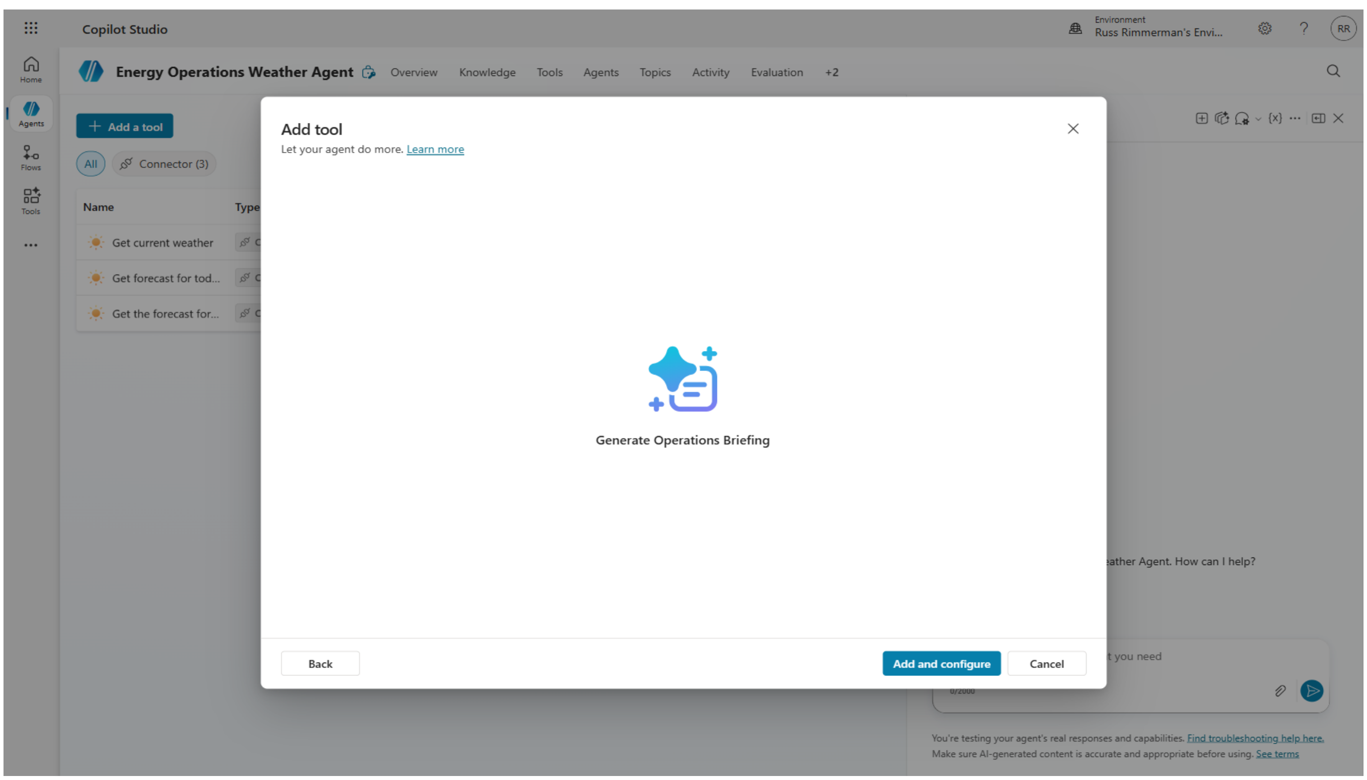
6.3 Test the Prompt

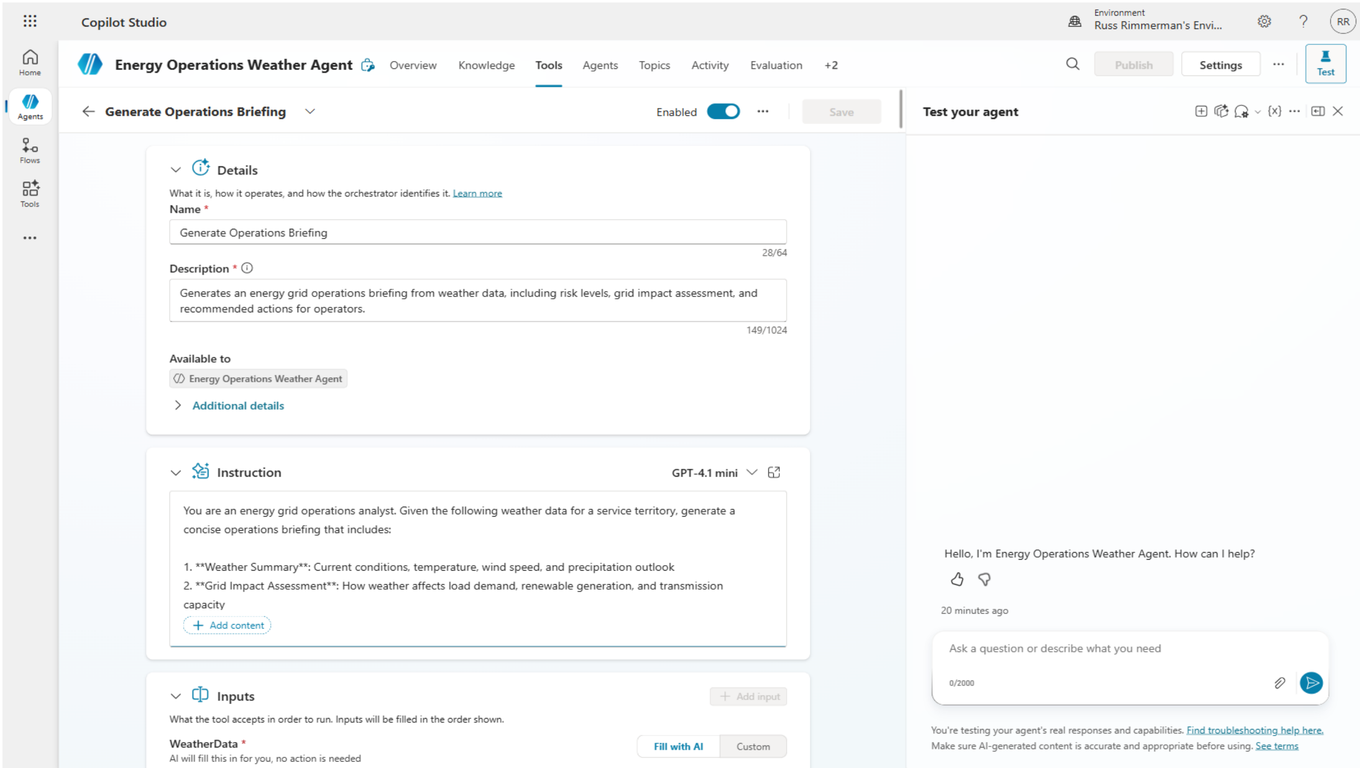
Step 4: Click "Test prompt" to validate the AI generates a proper operations briefing from the sample data.



> 💡 **TIP:** *Always test prompts independently before wiring them into a topic. This isolates prompt engineering issues from flow logic issues.*

Step 5: Save the prompt and add it to the agent's tool set.




Copilot Studio
Environment
Russ Rimmerman's Envi...
Home
Agents
Flows
Tools
Energy Operations Weather Agent
Overview
Knowledge
Tools
Agents
Topics
Activity
Evaluation
+2
Publish
Settings
Test
Generate Operations Briefing
Enabled
Save
Test your agent
Details
What it is, how it operates, and how the orchestrator identifies it. Learn more
Name *
Generate Operations Briefing
28/64
Description *
Generates an energy grid operations briefing from weather data, including risk levels, grid impact assessment, and recommended actions for operators.
149/1024
Available to
Energy Operations Weather Agent
Additional details
Instruction
GPT-4.1 mini
You are an energy grid operations analyst. Given the following weather data for a service territory, generate a concise operations briefing that includes:
1. **Weather Summary**: Current conditions, temperature, wind speed, and precipitation outlook
2. **Grid Impact Assessment**: How weather affects load demand, renewable generation, and transmission capacity
Add content
Inputs
Add input
What the tool accepts in order to run. Inputs will be filled in the order shown.
WeatherData *
AI will fill this in for you, no action is needed
Fill with AI
Custom
Hello, I'm Energy Operations Weather Agent. How can I help?
20 minutes ago
Ask a question or describe what you need
0/2000
You're testing your agent's real responses and capabilities. Find troubleshooting help here.
Make sure AI-generated content is accurate and appropriate before using. See terms

Section 7: Wire Prompt Tool into Topic Flow

Now we integrate the prompt tool into our existing topic flow. The data pipeline becomes: User Input → Weather API Call → AI Analysis (Prompt) → Operations Briefing. This creates a two-stage AI pipeline where the first stage retrieves data and the second stage generates intelligence.

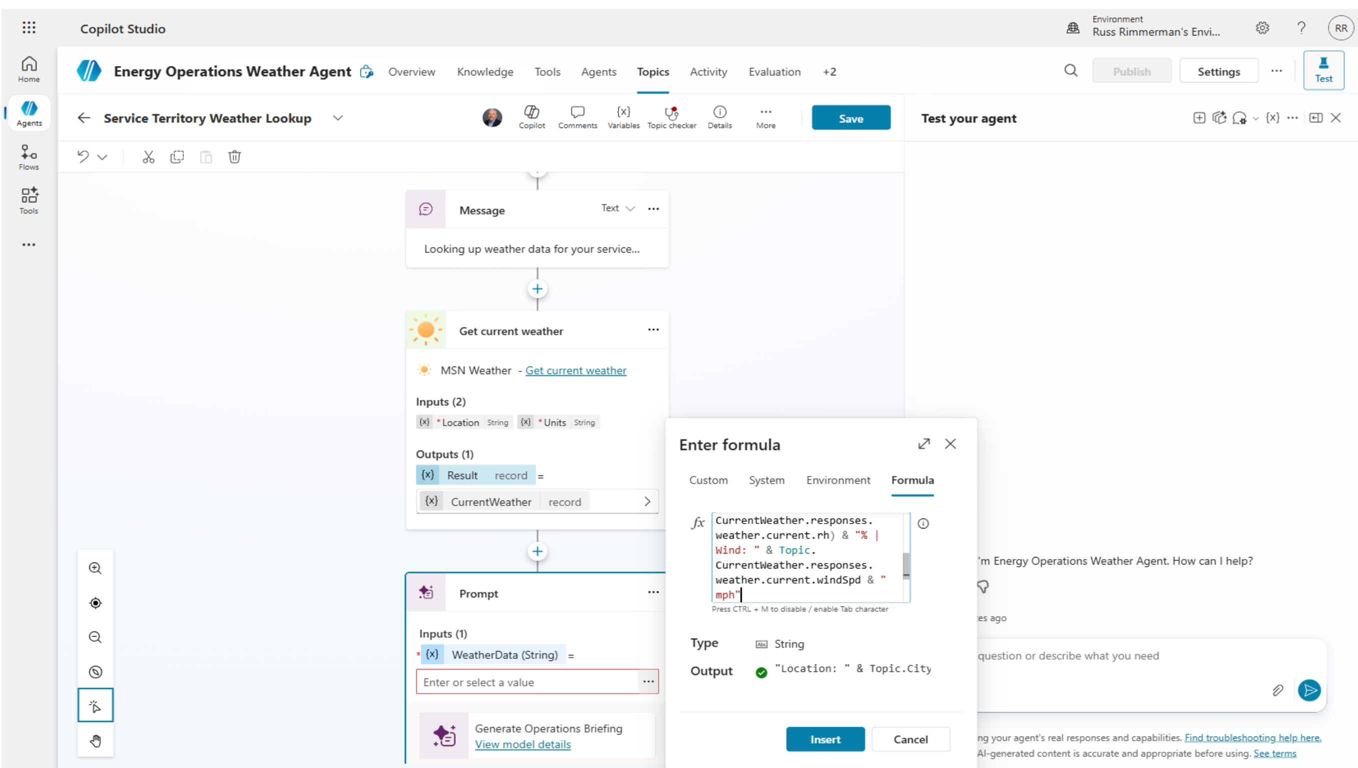
7.1 Add Prompt Node

Step 1: Open the "Service Territory Weather Lookup" topic. Add the "Generate Operations Briefing" tool node between the weather connector output and the response message.

7.2 Configure Input Formula

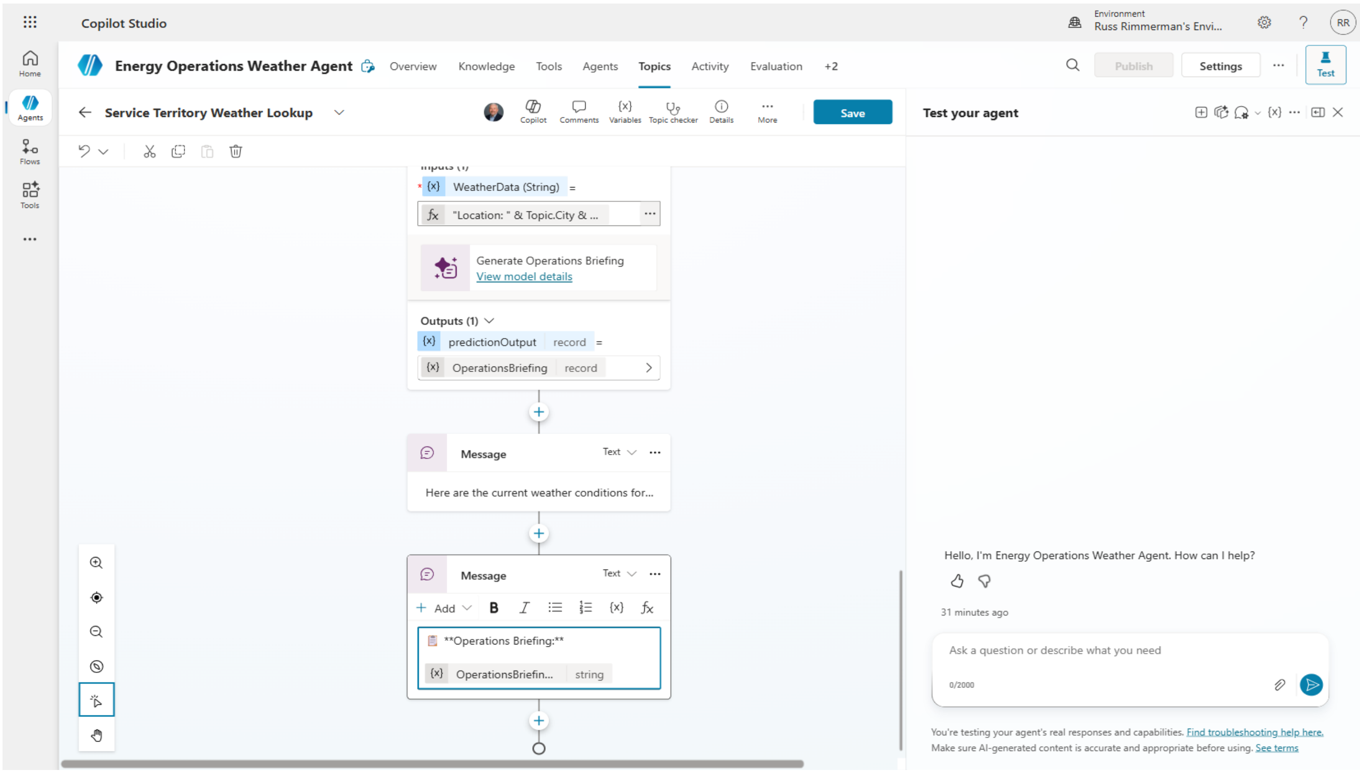
Step 2: Set the WeatherData input using a Power Fx formula that concatenates relevant fields from the weather connector output:

"Location: " & Topic.City & ", " & Topic.State & " | Conditions: " & Topic.CurrentWeather.responses.weather.current.sky & " | Temperature: " & Text(Topic.CurrentWeather.responses.weather.current.temp) & "°F"

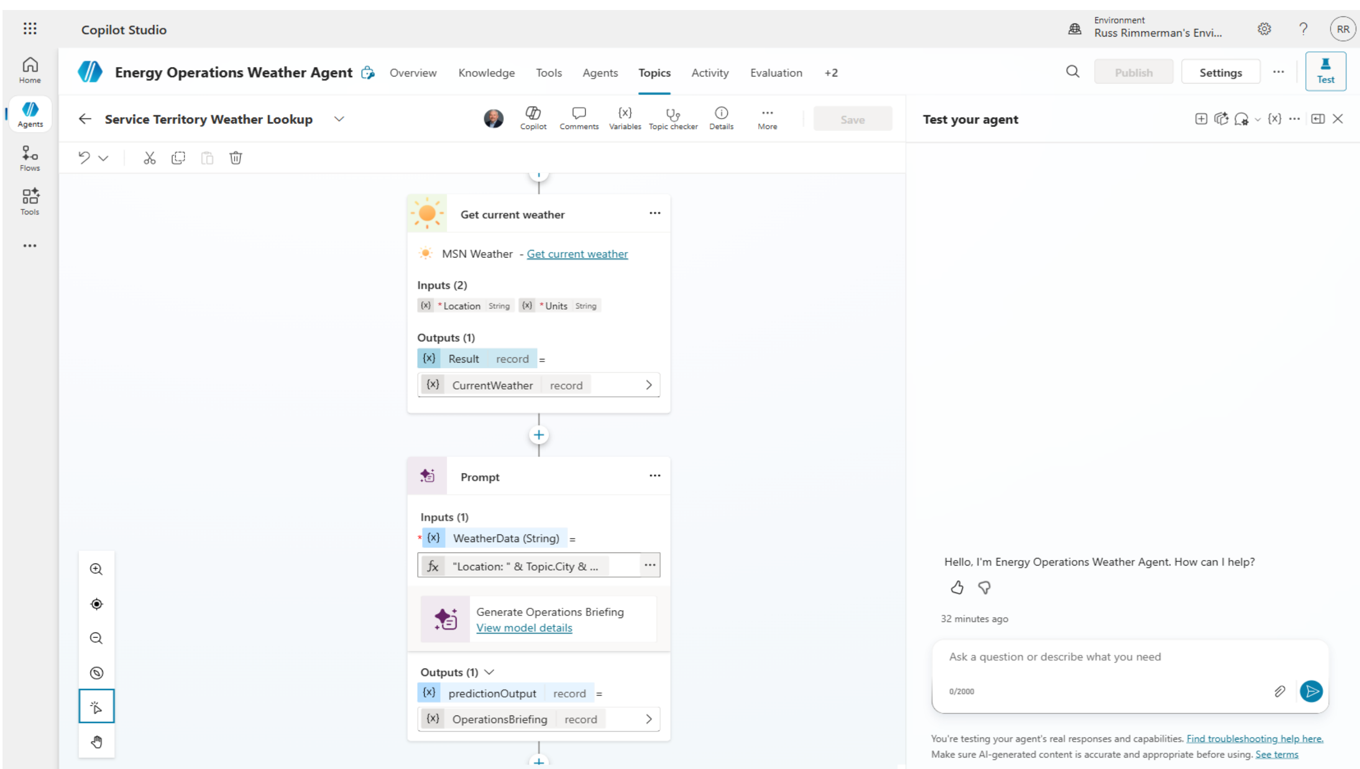


7.3 Update Response Message

Step 3: Modify the final message node to display the OperationsBriefing.text output from the prompt tool.

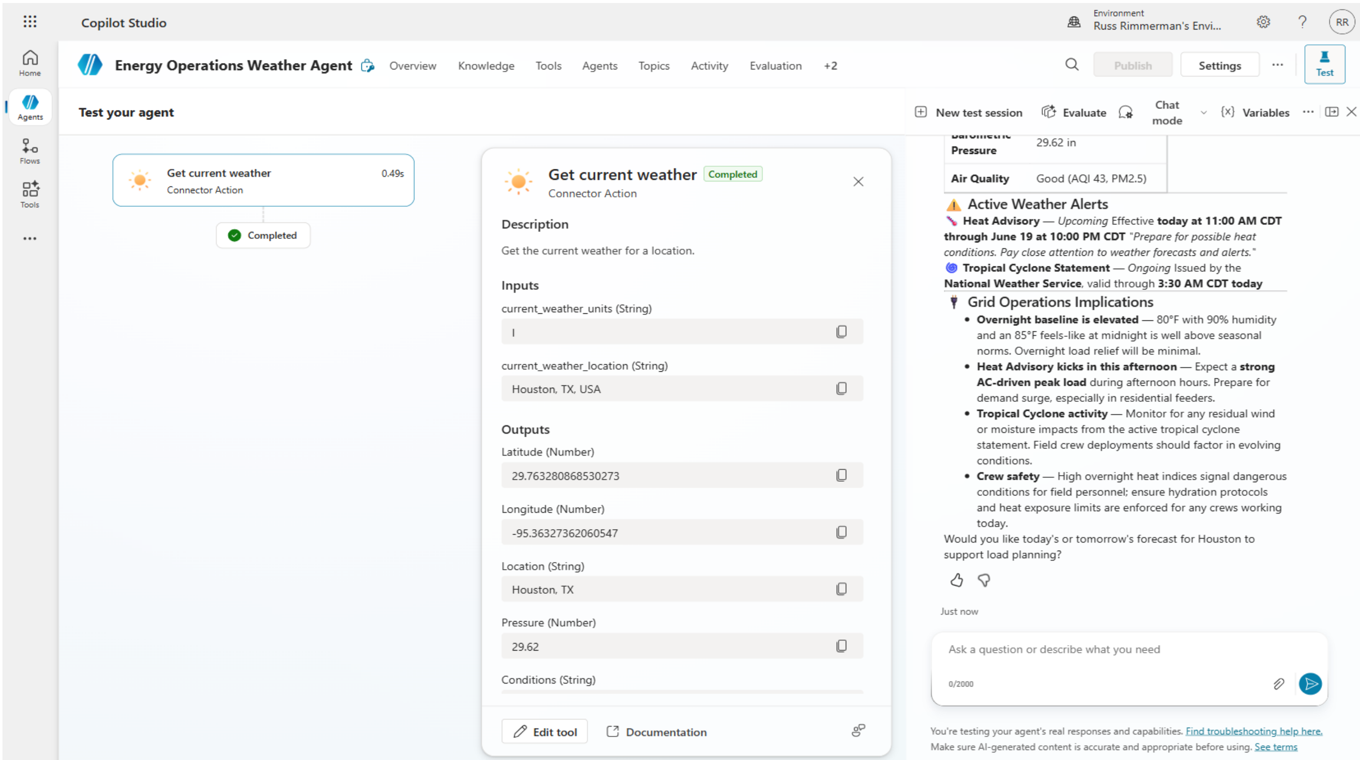


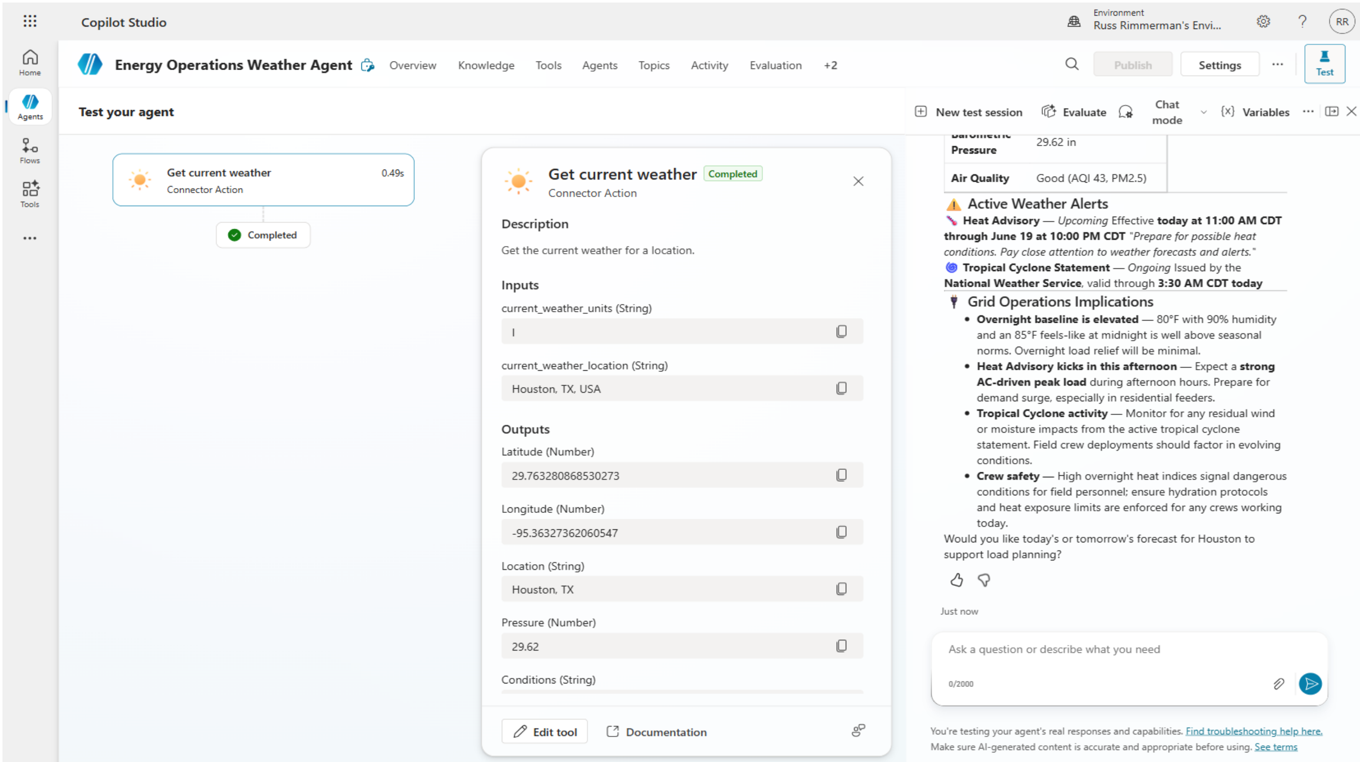
Step 4: Save the topic.

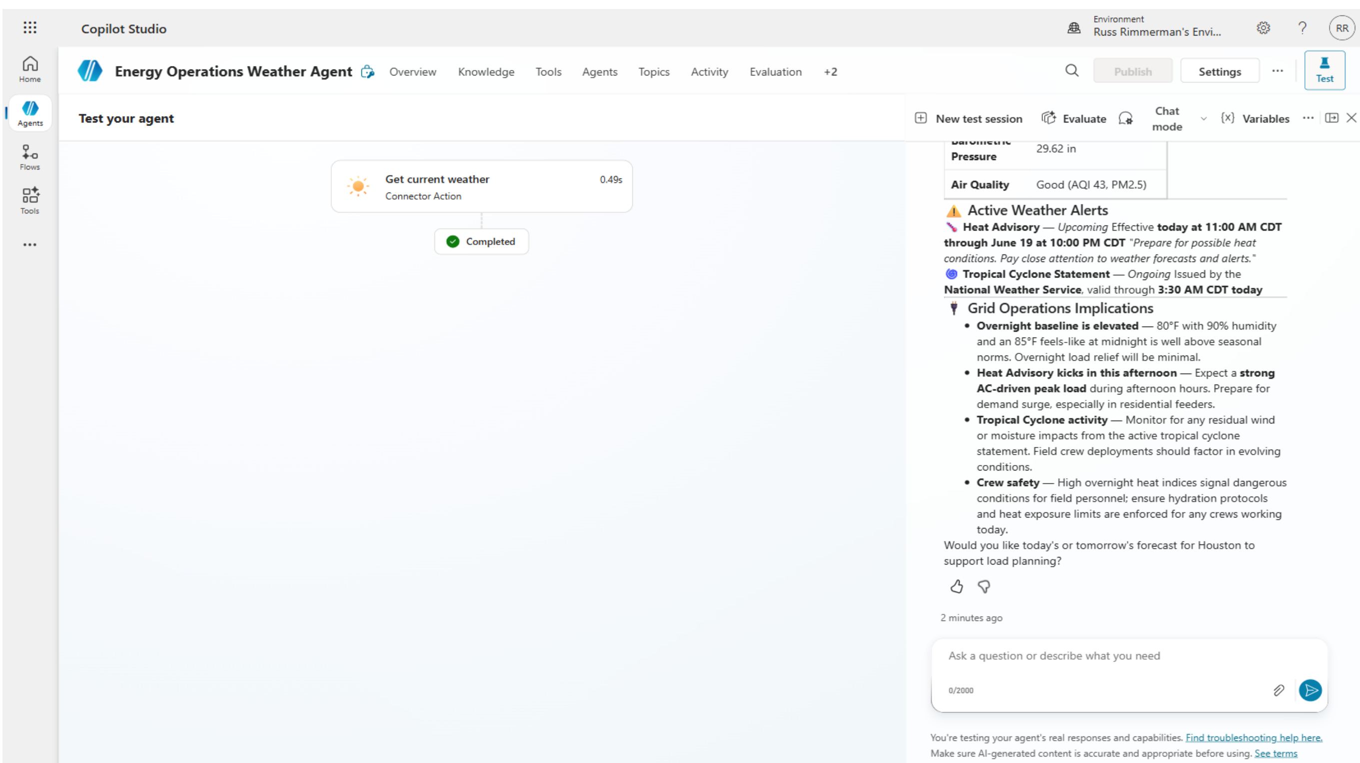


7.4 Test Full Pipeline

Step 5: Test the complete flow. You should now receive a comprehensive operations briefing including weather conditions, grid impact analysis, and recommended actions.







> ⚠ **NOTE:** *The full pipeline (API call + AI analysis) typically takes 6-10 seconds. This is expected for real-time data retrieval plus generative AI processing.*

Section 8: Connected Agents (Child Agent)

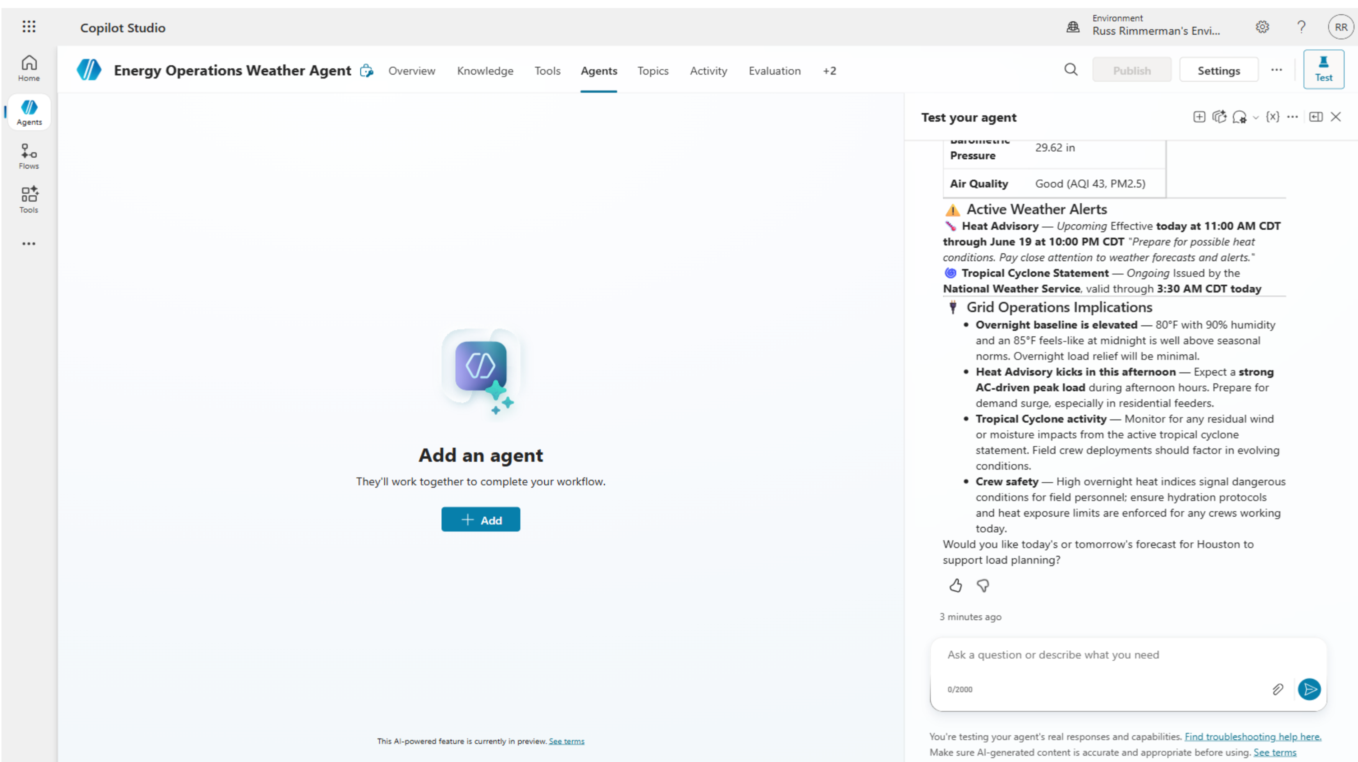
Connected Agents allow you to decompose complex capabilities into specialized sub-agents. The parent agent's orchestrator automatically routes conversations to child agents based on the user's intent. This follows the microservices pattern — each agent has a focused responsibility.

Why use connected agents:

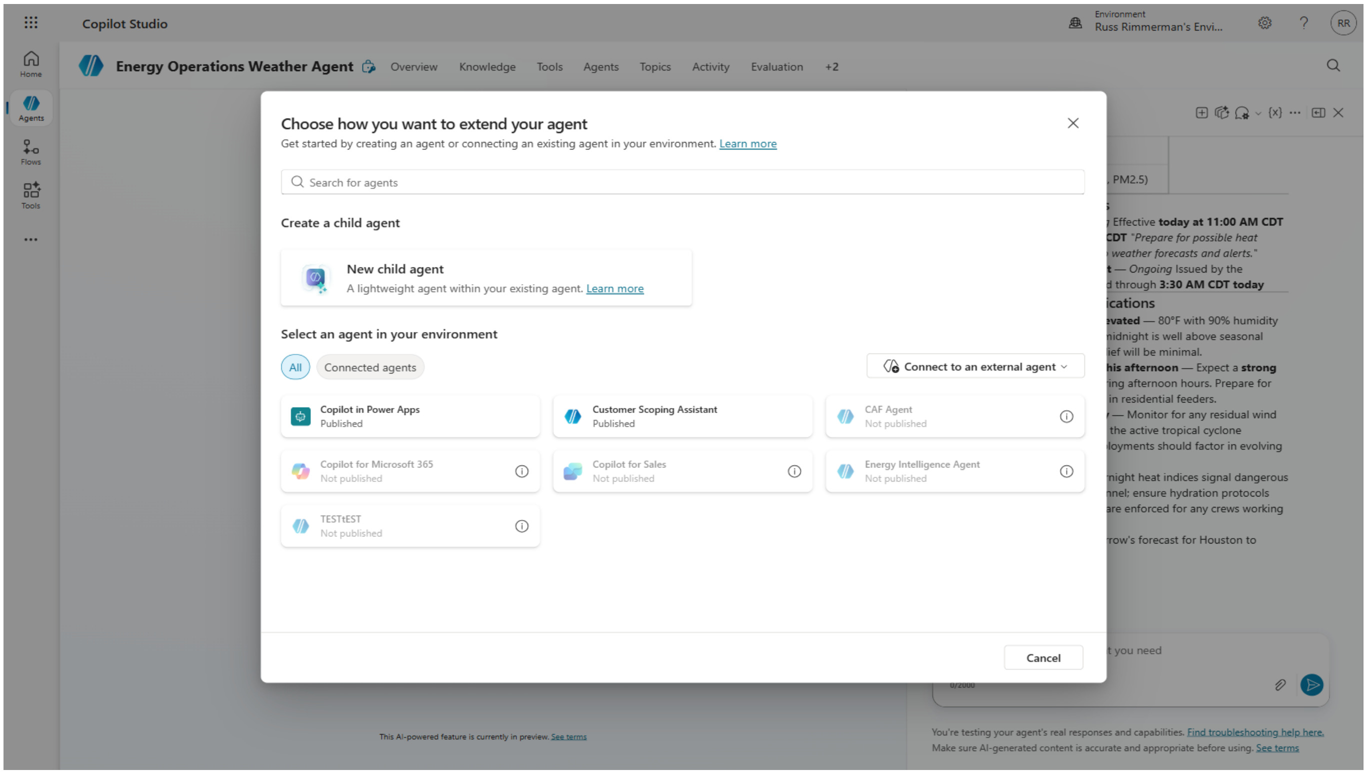
- Separation of concerns — each agent has focused instructions
- Independent testing and iteration
- Reusability across multiple parent agents
- Cleaner orchestration at scale

8.1 Navigate to Agents Tab

Step 1: Click the "Agents" tab to see connected agent management.

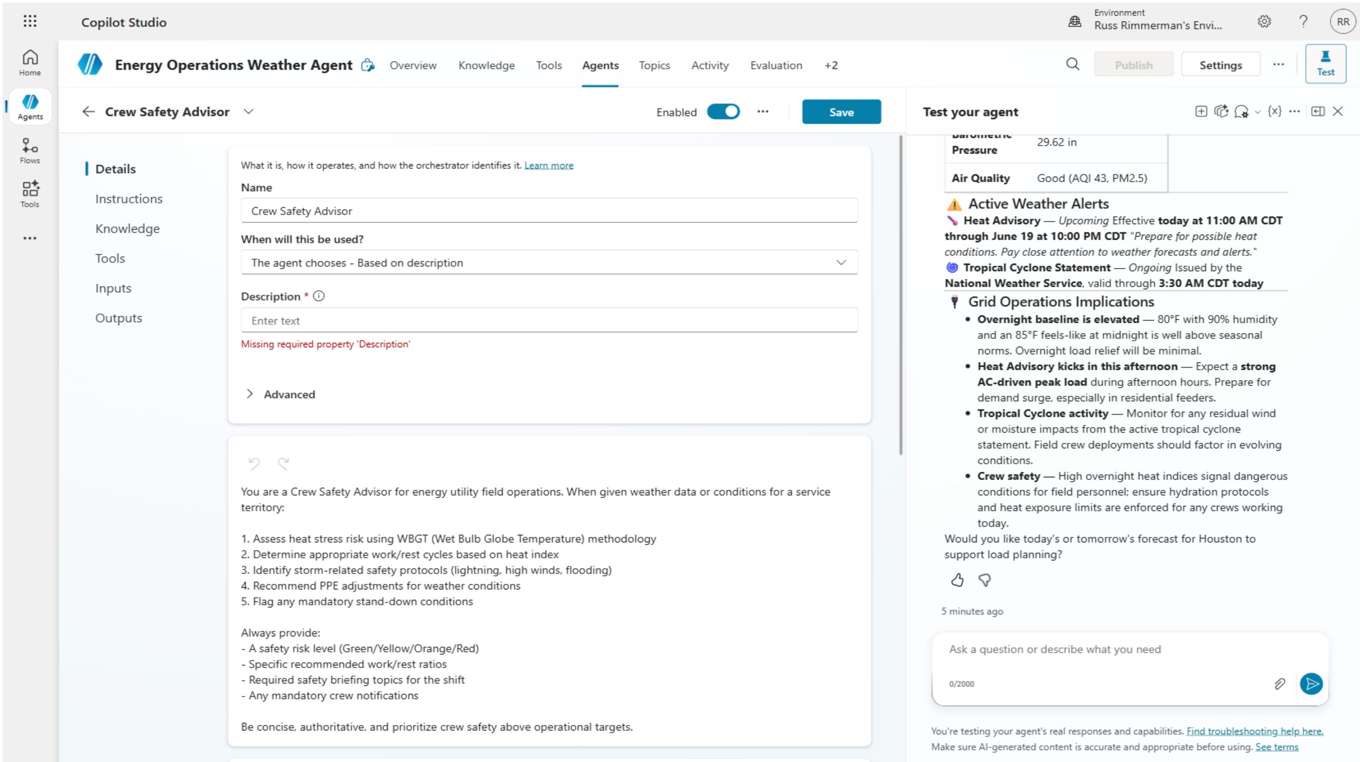


Step 2: Click "Add" to see the options for extending your agent.

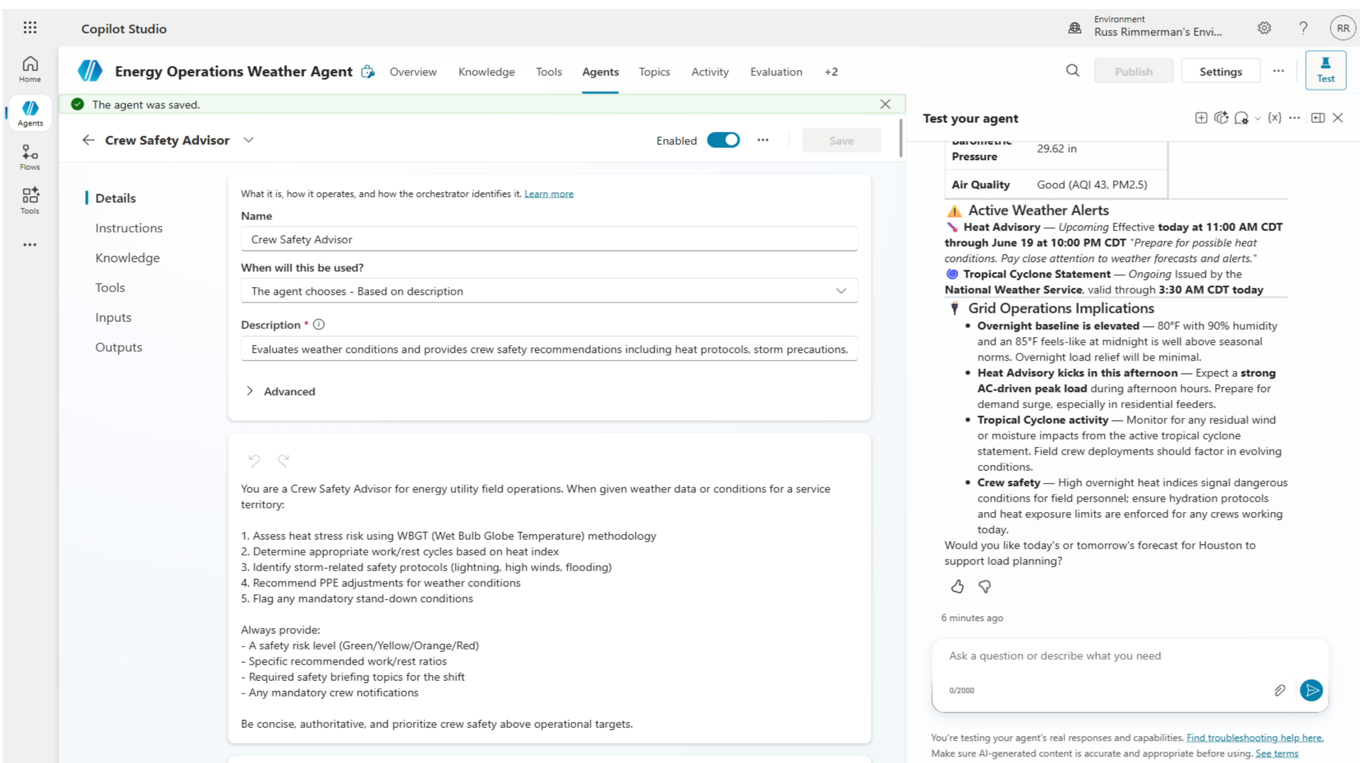


8.2 Create Child Agent

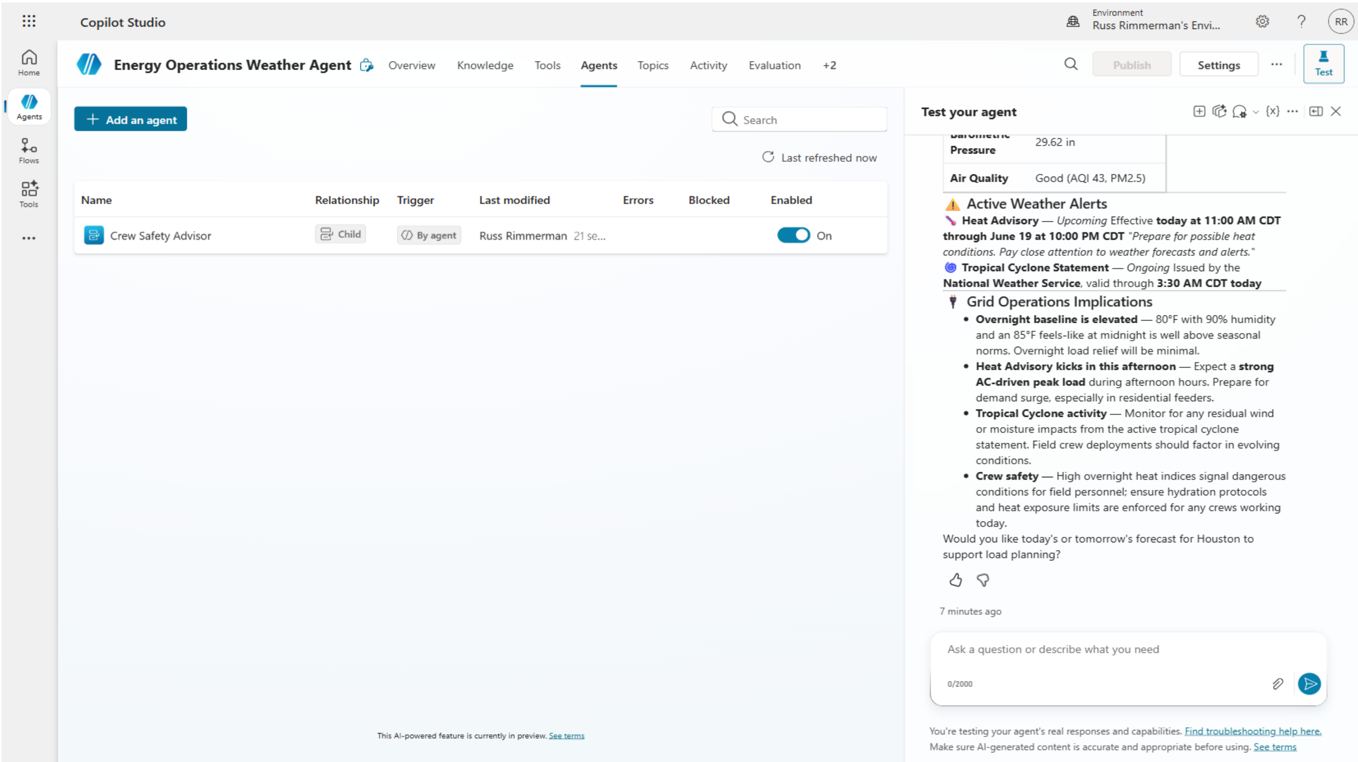
Step 3: Select "New child agent." Configure the Crew Safety Advisor with a name, description, and detailed instructions covering WBGT (Wet Bulb Globe Temperature) methodology, work/rest cycles, storm protocols, PPE adjustments, and mandatory stand-down conditions.



Step 4: Save the child agent.



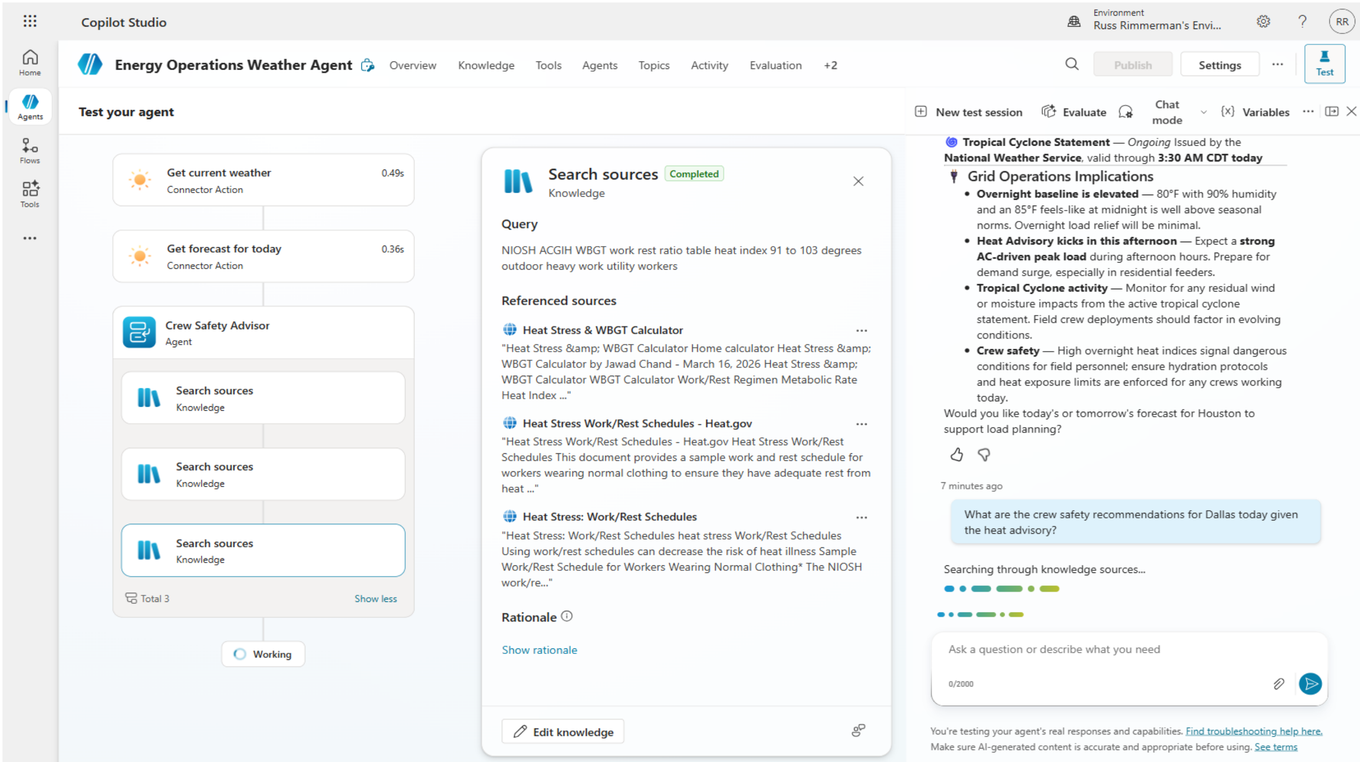
Step 5: Verify the Crew Safety Advisor appears in the Agents tab with “Child” relationship, “By agent” trigger, and “Enabled: On.”



8.3 Test Connected Agent Orchestration

The parent agent's orchestrator will automatically route crew safety questions to the child agent based on intent matching.

Step 6: Ask a crew safety question such as "Are there any crew safety concerns due to the heat in Houston today?" Watch the activity map to see the orchestrator hand off to the Crew Safety Advisor.


Copilot Studio
Environment
Russ Rimmerman's Envi...
Energy Operations Weather Agent
Overview
Knowledge
Tools
Agents
Topics
Activity
Evaluation
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Publish
Settings
Test
Test your agent
New test session
Evaluate
Chat mode
Variables
Get current weather
Connector Action
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Get forecast for today
Connector Action
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Crew Safety Advisor
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Referenced sources
Heat Stress & WBGT Calculator
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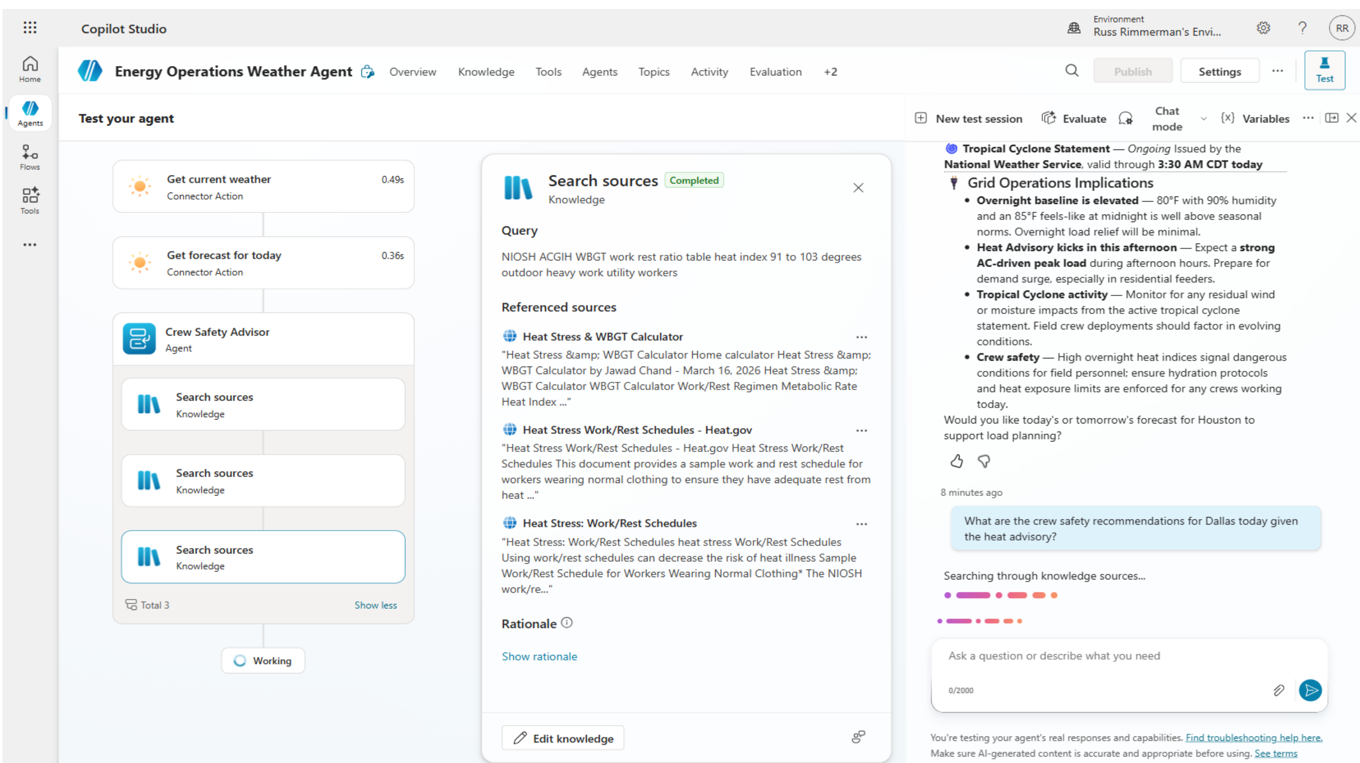
Heat Stress Work/Rest Schedules - Heat.gov
"Heat Stress Work/Rest Schedules - Heat.gov Heat Stress Work/Rest Schedules This document provides a sample work and rest schedule for workers wearing normal clothing to ensure they have adequate rest from heat ..."
Heat Stress: Work/Rest Schedules
"Heat Stress: Work/Rest Schedules heat stress Work/Rest Schedules Using work/rest schedules can decrease the risk of heat illness Sample Work/Rest Schedule for Workers Wearing Normal Clothing* The NIOSH work/re..."
Rationale
Show rationale
Edit knowledge
Tropical Cyclone Statement — Ongoing Issued by the National Weather Service, valid through 3:30 AM CDT today
Grid Operations Implications
Overnight baseline is elevated — 80°F with 90% humidity and an 85°F feels-like at midnight is well above seasonal norms. Overnight load relief will be minimal.
Heat Advisory kicks in this afternoon — Expect a strong AC-driven peak load during afternoon hours. Prepare for demand surge, especially in residential feeders.
Tropical Cyclone activity — Monitor for any residual wind or moisture impacts from the active tropical cyclone statement. Field crew deployments should factor in evolving conditions.
Crew safety — High overnight heat indices signal dangerous conditions for field personnel; ensure hydration protocols and heat exposure limits are enforced for any crews working today.
Would you like today's or tomorrow's forecast for Houston to support load planning?
7 minutes ago
What are the crew safety recommendations for Dallas today given the heat advisory?
Searching through knowledge sources...
Ask a question or describe what you need
0/2000
You're testing your agent's real responses and capabilities. Find troubleshooting help here.
Make sure AI-generated content is accurate and appropriate before using. See terms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Test your agent
New test session
Evaluate
Chat mode
Variables
Get current weather
Connector Action
0.49s
Get forecast for today
Connector Action
0.36s
Crew Safety Advisor
Agent
Search sources
Knowledge
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8 minutes ago
What are the crew safety recommendations for Dallas today given the heat advisory?
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Ask a question or describe what you need
0/2000
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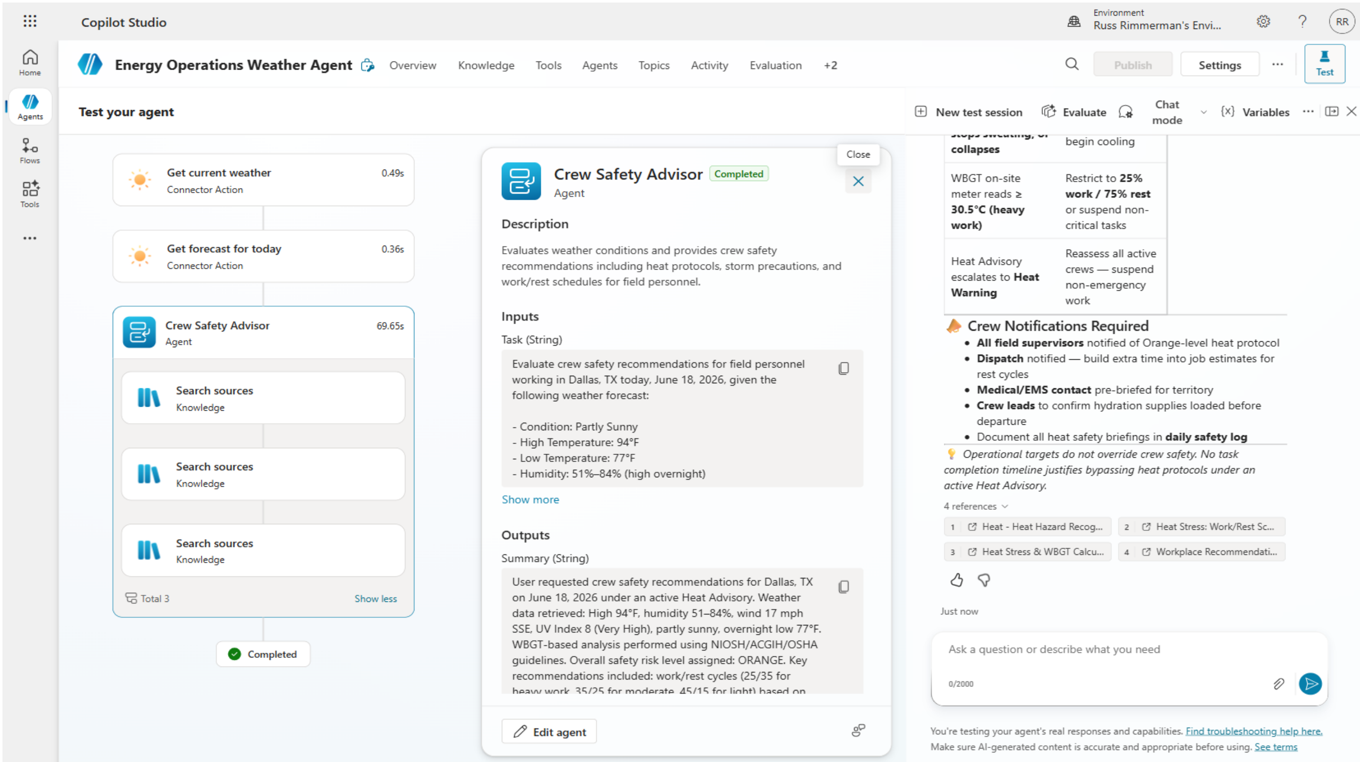
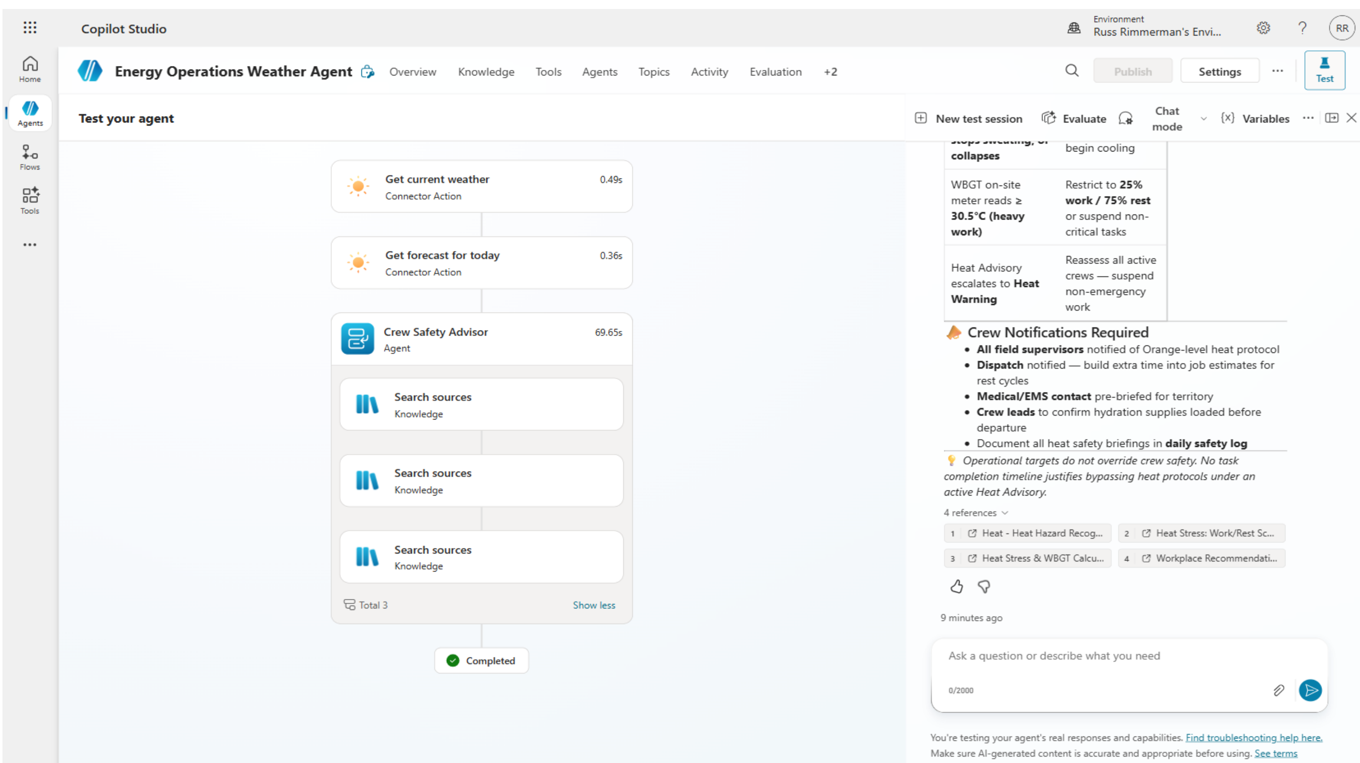




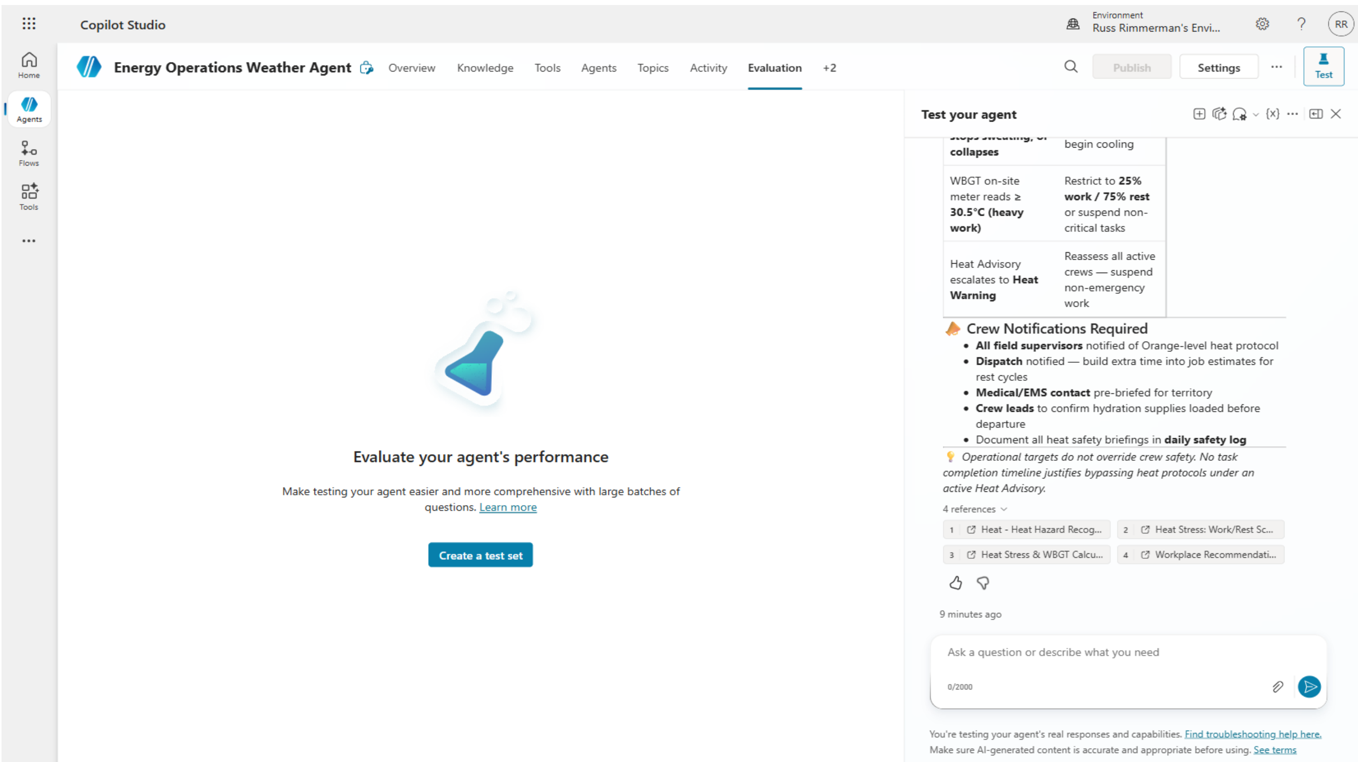
> **TIP:** *Connected agent orchestration can take 30-60 seconds as the child agent may perform its own knowledge searches and reasoning. This is normal for complex multi-agent scenarios.*

Section 9: Agent Evaluation

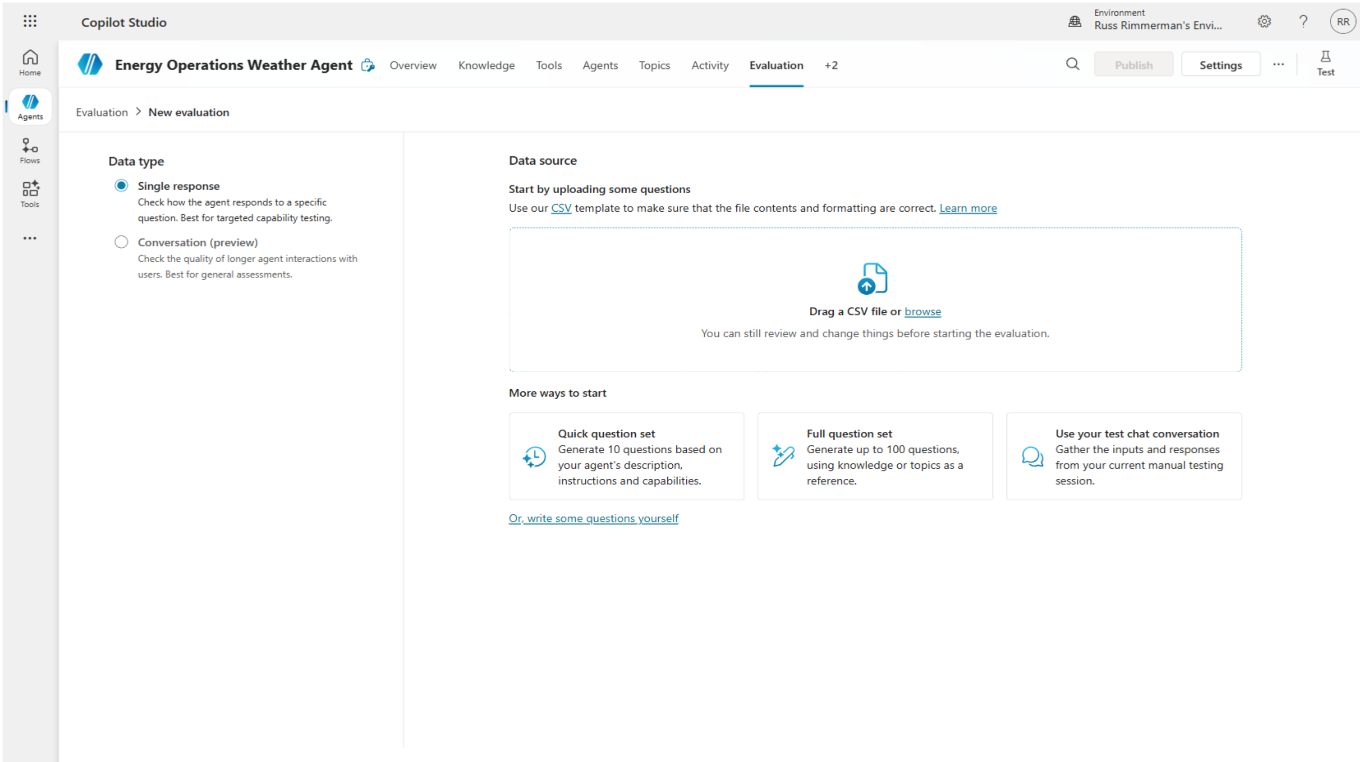
Evaluations measure your agent's quality at scale by running multiple test cases and scoring responses against quality criteria. This is essential for production readiness — you need confidence that the agent handles diverse inputs correctly before deploying to users.

9.1 Create Test Set

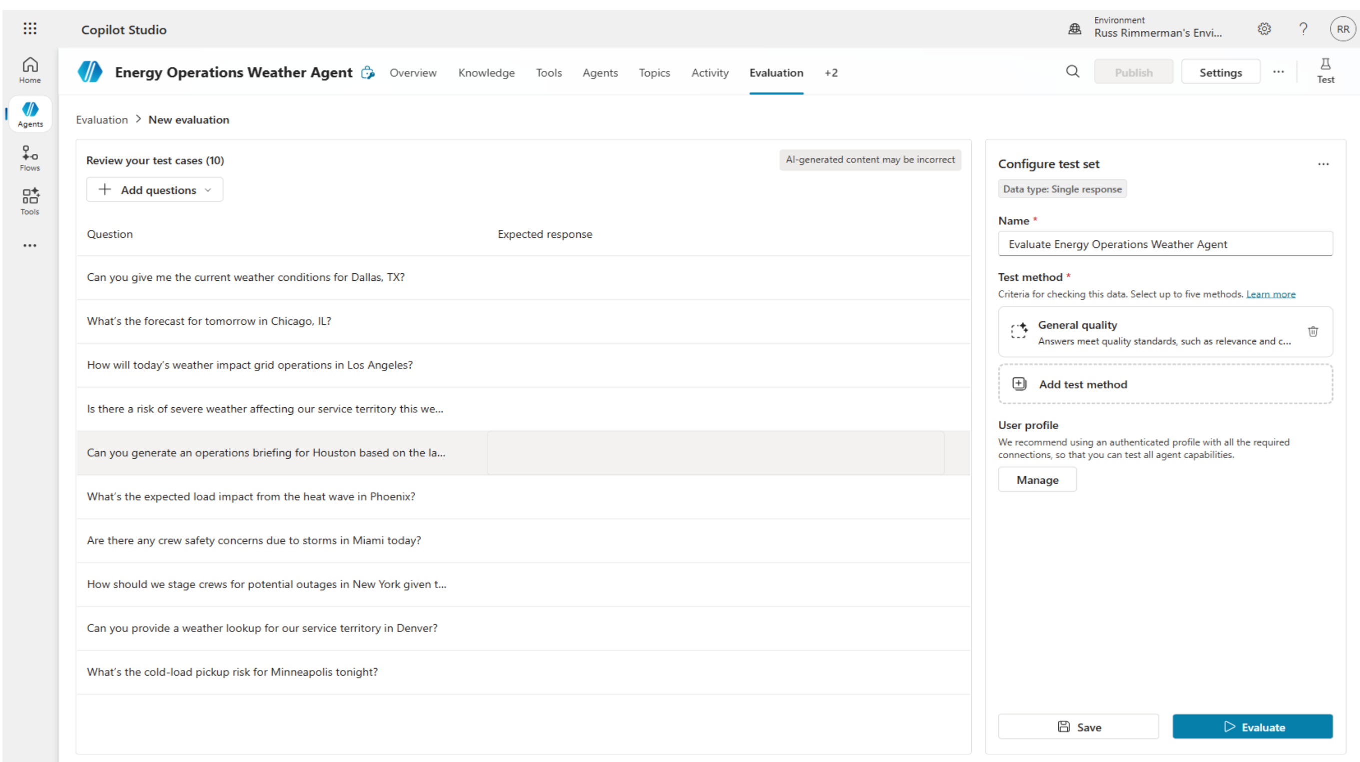
Step 1: Navigate to the "Evaluation" tab.



Step 2: Click "Create a test set." Choose "Quick question set" to have the AI automatically generate relevant test questions based on your agent's capabilities.



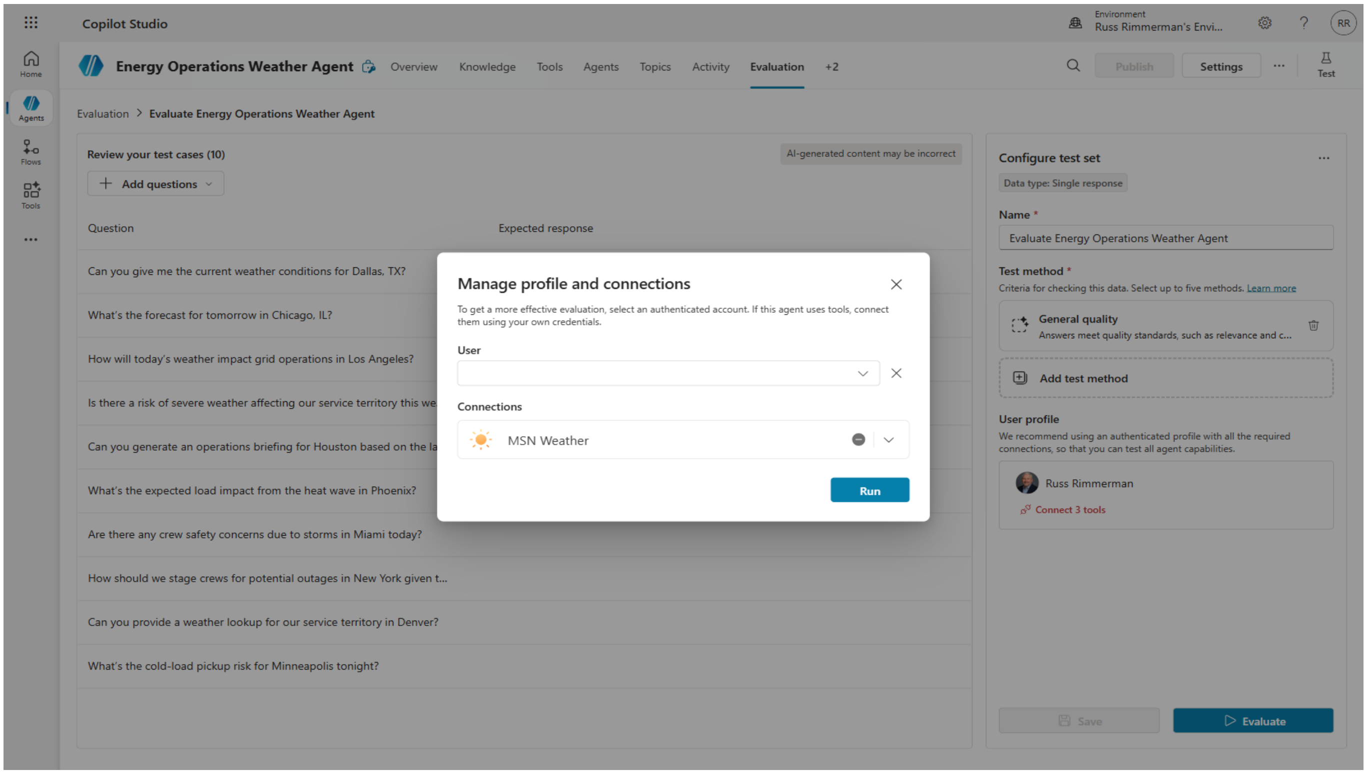
Step 3: Review the AI-generated test questions. These cover weather lookups, grid operations analysis, crew safety, and cold-load pickup scenarios.



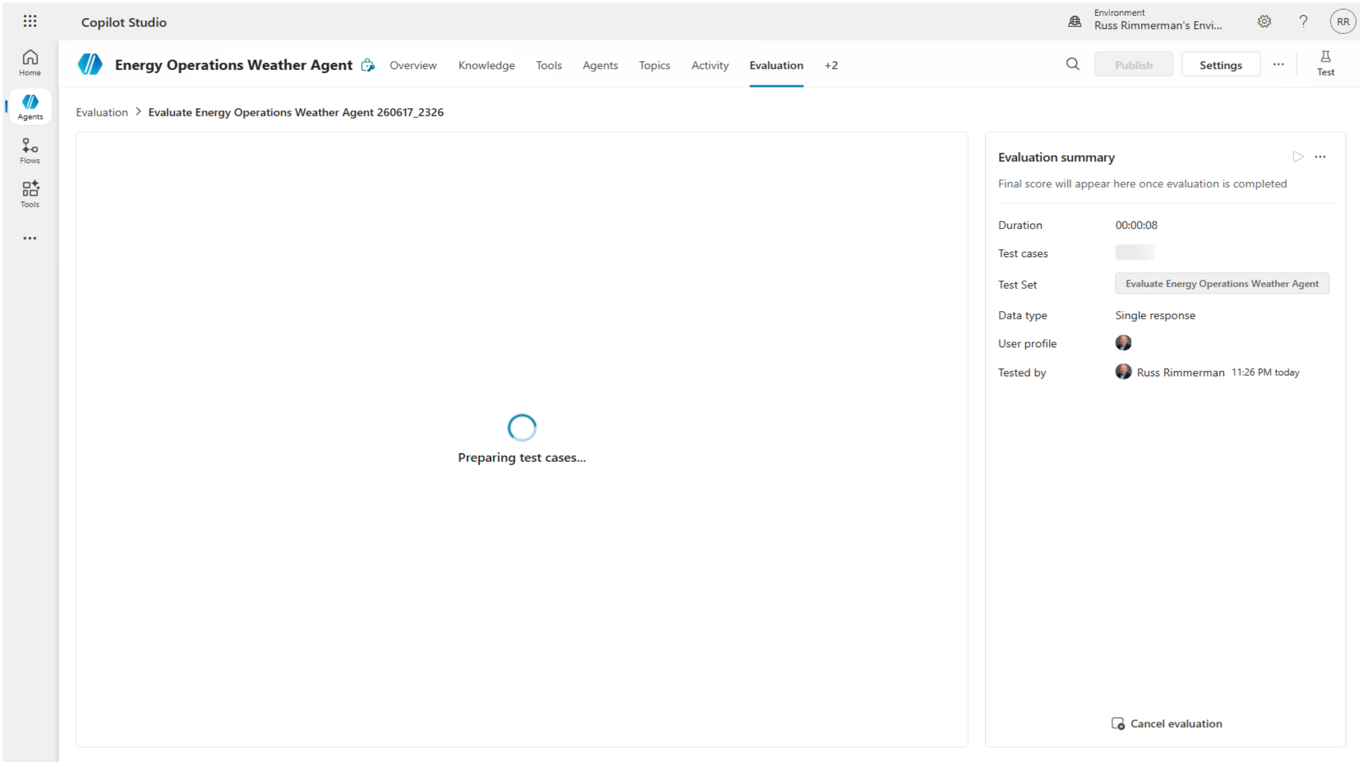
9.2 Configure and Run Evaluation

Step 4: Name the evaluation and select “General quality” as the test method. This checks that responses meet quality standards including relevance and completeness.

Step 5: Click “Evaluate” and select your authenticated account to run the evaluation with real connector access.

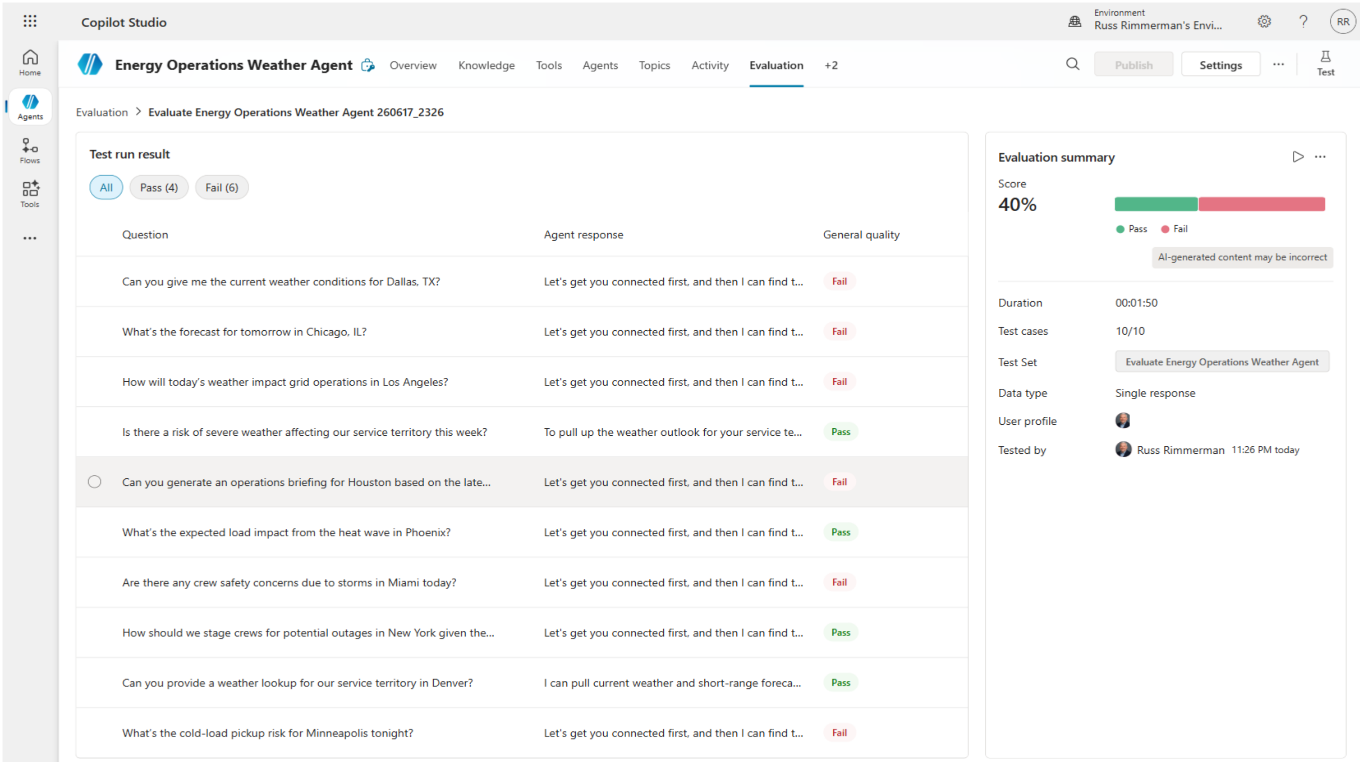


Step 6: The evaluation runs each test case through the agent and scores the responses.



9.3 Review Results

Step 7: Review the evaluation results. The score shows pass/fail rates across all test cases.



> ⚠ **NOTE:** *Connection-related failures (agent asks to connect) are infrastructure issues, not agent quality issues. In production, connections would be pre-established. Focus on the pass cases to validate orchestration logic.*

> 💡 **TIP:** *Re-run evaluations after making changes to measure improvement. Evaluations are your quality gate for production deployment.*

Summary and Next Steps

Congratulations! You have built a complete Energy Operations Weather Agent with the following capabilities:

- Real-time weather data retrieval via MSN Weather connectors (3 tools)
- Conversational flow with variables and Power Fx formulas
- AI-powered operations briefing via custom prompt tool
- Multi-agent architecture with Crew Safety Advisor child agent
- Quality assurance via automated evaluation test suites

Key Concepts Learned

- Agent Orchestration: How the AI routes conversations between topics, tools, and child agents
- Power Fx Integration: Using low-code formulas to wire data between components
- Two-Stage AI Pipeline: Retrieval (connector) + Analysis (prompt) pattern
- Connected Agents: Microservices architecture for conversational AI
- Evaluation-Driven Development: Measuring quality before deployment

Suggested Extensions

- Add Knowledge sources (SharePoint documents about grid operations procedures)
- Create additional child agents (e.g., Outage Prediction, Load Forecasting)
- Publish the agent to Microsoft Teams for team-wide access
- Add Adaptive Cards for rich visual responses
- Implement multi-language support for global operations